scientific reports

OPEN

An advanced skin lesion segmentation and classification framework using deep learning strategies

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Skin cancer is a deadly kind of cancer that grows rapidly and produces life-threatening issues within six weeks from the initial stage. Accurate analysis is needed for observing both the malignant and benign skin lesions that are more complicated to ensure the essential treatment for the individuals. The best solution to perform automatic skin lesion categorization is to employ deep learning systems that are trained from a huge amount of benchmark data. Moreover, melanoma is referred to as a crucial skin cancer that badly affects entire human health and also their life. In this case, early-stage melanoma diagnosis is more complicated due to the presence of color changes in the same lesion. Hence, to overcome these complications in the traditional techniques, a novel automated skin lesion segmentation model is required to provide effective diagnosis results. The collected images are applied to the preprocessing stage to enhance quality. The pre-processed images are given to the segmentation stage by utilizing an Adaptive Layer-based Visual Transformer with UNet (AL-VTransUNet), where the variables are tuned with Improved Random Parameter-based Galactic Swarm Optimization (IRP-GSO). Ultimately, the segmented images are subjected to the Dilated DenseNet with a Multi-Head Attention mechanism (DD-MHA) for classifying the outcomes. Here, the variable is also tuned with the support of the same IRP-GSO algorithm for improving classification effectiveness. Thus, the efficiency of the recommended framework is examined using existing optimization and classification approaches.

Keywords Skin lesion segmentation and classification, Improved random parameter-based galactic swarm optimization, Dilated densenet with multi-head attention mechanism, Adaptive layer-based visual transformer with UNet

Melanoma is a deadly kind of skin malignancy that has greatly reduced the survival rate of individuals in recent years[1]. Researchers have found that the occurrence rate of melanoma has been faster in past years[2], and also, the death rate of the individuals is increased if it is not identified in the initial stage[3]. The primary cause of skin cancer is widespread exposure to Ultraviolet (UV) radiation from sunlight, which can damage skin cells that leads to cancerous development. It is more destructive compared to other types of cancers. Some people may not identify the early signs of skin cancer, leading to delayed diagnosis. Hence, early recognition of melanoma is highly essential[4]. The diagnostic tool supports the professionals in detecting cancer in the beginning phases, and this recognition helps the patients to recover from this deadly cancer. Dermoscopy is termed the non-invasive model, which is utilized to identify melanoma[5]. They perform visual analysis in pigmented skin lesions by applying a gel substance over the affected regions. They execute several inspections with a magnification tool named dermoscopy that amplifies the skin image up to 100 times[6]. The dermoscopy executes a detailed analysis of the subsurface skin structure along with diagnosis for effective clinical applications of the lesion analysis[7].

Epiluminescence Microscopy (ELM) is a non-invasive analyzing approach, that is used to detect skin lesions. ELM permits visualization in the skin subsurface by exploring the aspects of lesions, like texture and color. Moreover, ELM analysis enhances the skin lesion identification rate regarding naked-eye observation and achieves a higher accuracy rate[8]. The accuracy of ELM findings is based on the knowledge of a dermatologist[9]. The CAD systems are utilized to examine benign skin lesions to restrict the development of malignant lesions. Further, researchers perform active research over computer-based observations and diagnosis of melanoma[10].

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Advanced techniques are implemented for the recognition of malignancy in skin images that process the images acquired by digital cameras[11]. The dermoscopic images have uniform illumination, and also they are available with a higher contrast rate. The non-dermoscopic clinical images used for the analysis are commonly available[12]. Standard skin digital images are highly complex to analyze as they include artefacts like uneven reflection, illumination, and availability of noise. These kinds of artefacts affect the performance of the basic segmentation models[13]. So, intelligent segmentation approaches are developed for analyzing non-dermoscopic digital images, and some approaches use the color information presented in the single channel or efficient channels based on Principle Component Analysis (PCA)[14]. Most of the segmentation approaches utilize preprocessing steps to deal with the artefacts and illumination effect[15].

Moreover, deep learning approaches offered better outcomes in medical imaging applications and pattern recognition models[16]. Here, Convolutional Neural Networks (CNN) are commonly utilized in deep structure architecture inspired by the human visual cortex. Even though, CNN is implemented in various imaging applications[17]. Most CNN configurations are designed to identify the salient area in the image, and the entire super-pixel is labeled by global and local context. Some investigative works use conversion techniques to identify the affected regions and separate them from backdrop pixels. These techniques change the input RGB images into other color spaces[18]. The supervised learning techniques execute skin lesion segmentation by performing network training. The dermoscopic image segmentation procedure is performed using Fully Convolutional Networks (FCN) with several enhancements in the unusual loss function like Jaccard distance[19]. Later, the imbalance issues are resolved to provide effective results over skin lesion classification.

Motivation of the research work: The recognition of melanoma is needed to provide effective treatment to the patients, which improves the quality of their lives. If melanoma is identified in the initial phases, then the survival rates of humans are improved effectively. However, the visual resemblance between malignant and benign generates more complexity in detecting melanoma[20]. Thus, the identification of melanoma became more complicated and tedious to the researchers. In the current scenario, more imaging techniques are developed to analyze the dermoscopy image that allows the visual feature of skin region by tiny outstanding components. They are widely utilized for demography analysis and also improve the detection rate[21]. Because of the requirement of human vision, melanoma detection in skin lesions is more challenging. So these kinds of complications presented in the identification process need to be resolved, so professionals developed automated SLSC methods to increase the efficacy of the skin lesion categorization rate. The automated skin lesion classification process utilized four major procedures for performing effective analysis[22]. Identifying the occurrence of melanoma and lesions is the essential step. However, they became more complicated due to dissimilarity presented in the location, shape, and size in skin lesion images. Moreover, the poor dissimilarity rate in the image leads to discrimination in the nearby tissue cells[23]. Other constraints that affect the better skin lesion classification rate are hair, ruler marks, color illumination, air bubbles, and blood vessels. So, it is crucial to resolve the limitations available in the conventional skin lesion detection model, so an innovative framework for effective SLSC is designed based on deep learning models.

Different contributions of the developed SLSC framework are detailed as follows.

- To propose an inventive SLSC technique with deep structured architecture to obtain a better skin lesion classification rate between the patient information in the beginning stages, assist clinical psychologist in making more accurate diagnoses, leading to better treatment plans, and help in the early detection of potential skin diseases.
- To construct AL-VTransU-Net gives helpful details about the size, shape, and characteristics of skin lesions, helping doctors to provide appropriate treatment plans. The optimal tuning of parameters of TransU-Net using IRP-GSO improves the dice coefficient rate, thus providing more effective results over classification.
- To develop an innovative skin lesion classification framework named DD-MHA can supervise the development of skin diseases, calculate the effectiveness of treatments, and accurately identify different types of skin lesions, aiding in the diagnosis of various skin conditions by modifying the parameters of dilated DenseNet for increasing the rate of precision and accuracy in the skin lesion framework with IRP-GSO.
- To implement a novel heuristic model IRP-GSO to tune the parameters of TransU-Net like batch size, n-layer, learning rate, and dropout and also the parameters of dilated DenseNet in the designed AL-VTransU-Net model for improving and accurately assessing the efficacy of the analysis.

The rest of the portions of the developed SLSC model are discussed below. Diverse baseline techniques are associated with SLSC approaches are described in phase II. An innovative skin lesion segmentation framework with a U-Net structure is explained in Phase III. An innovative skin lesion classification framework using dilated DenseNet and attention mechanism with the improved heuristic method is detailed in phase IV. Structural representation of the suggested SLSC method with deep structure models and the dataset utilized for the experimentation are illustrated in Phase V. Various analyses and observations performed over the developed SLSC with classical optimization method and classifiers are presented in Phase VI. The overall conclusions about the developed SLSC system are detailed in Phase VII.

Literature survey

Related works

Several methods were carried out for segmentation, classification and identification of skin lesions by adopting deep learning, and machine learning techniques. The literature review section briefly elaborates on in-depth insight of techniques with prior works based on skin lesion. The depth analysis in brain tumor classification is listed below. Xie et al.[24] generated a Mutual Bootstrapping-based Deep CNN (MB-DCNN) framework for identifying the skin lesion. Here, the segmentation and categorization network has transferred the data and eased the process by bootstrapping the model. To emphasize the classification outcome, the multi-view-based model effectively generated the discriminative data of multiple images with a weighted model by Bian et al.[25]. In several works, the weighting criterion was developed for a good solution according to texture feature analysis in the input images based on Akram et al.[26]. A highly dependable technique for efficient skin cancer finding using dermoscopic images was presented. Here, essential features for the analysis were collected using the Grasshopper Optimization Algorithm (GOA), and the suggested technique offered a better classification according to Thapar et al.[27]. The ensemble models by Myat et al.[28] for the skin lesion classification process using dermoscopic images were evaluated to strengthen the classification performance. Different observations executed on the developed system over the traditional skin lesion classification model obtained better lesion categorization rates.

For segmentation, active counter-based models were utilized among the skin and lesions to examine the lesion curve boundaries. So, the CAD technique was adopted for the effective identification of melanoma using dermoscopic images by Riaz et al.[29]. Specifically, the consideration of the U-Net model by Ramadan and Aly[30] modifies the input count to increase the efficacy rate and attain a better efficacy rate than the established methods. Navarro et al.[31] suggested a skin lesion segmentation technique with super-pixel models, and they secured an effective performance than the traditional approach. The recommended technique effectively operated over skin lesion images and captured the essential regions with diverse time scales. Sikkandar et al.[32] suggested the top hat filtering model, and also grab-cut technique to execute efficient segmentation in the pre-processed image. Kumar et al.[33] promoted an efficient deep-learning network for skin lesion segmentation and identification process. A Wiener filter was performed for preprocessing and a UNet + + model was suggested for skin lesion segmentation. End-to-end framework for SLSC was improved using the MobileNet by Sulthana et al.[34]. In addition, the image segmentation and extraction of features were done using a Gaussian filtering algorithm. Diverse skin color quantification methods by Bencevic et al.[35] were suggested for skin lesion segmentation against darker-skinned individuals.

For performing effective tumor analysis, the MagNet model was performed using histopathological images for maximizing the multi-magnification feature maps according to Iqbal et al.[36]. Iqbal et al.[37] have presented the adaptive self-learning in medical image analysis and classification model. Here, the developed framework has the ability to explore the image modality, preprocessing and feature extraction from both manual and CNN-based pertinent features. Iqbal et al.[38] have suggested the Adaptive Learning-based CAM (Adaptive-CAM) model to provide better connectivity among the action map and network predictions. The diverse deep learning models have improved the performance by analyzing the activation function, optimizing the hyperparameters, momentum and loss function in regards to Iqbal et al.[39]. To enhance the interpretability, the Hybrid Parallel Fuzzy Convolutional Neural Network (HP-FCNN) model developed by Iqbal et al.[40] has the ability to minimize the noise and hazy borders in Magnetic Resonance Imaging (MRI). Considering CNN and Local Binary Patterns (LBP) model by Iqbal et al.[41] is used to emphasize the classification performance by suggesting publicly available datasets like ISIC 2017, 2018 and 2019 (HAM10000). An enhanced deep learning-based strategy for classifying chest X-ray images has been utilized with pre-trained models by the author Arshad et al.[42], which enhances the clinical diagnostics in chest X-rays.

Problem statement

Recently, skin tumors have been regarded as the common natural malignancies to affect people worldwide. Several pros and cons of the established SLSC model are offered in Table 1. To resolve these demerits in the conservative model, the recommended SLSC framework is developed. The motivation of the SLSC approaches is as follows.

- The traditional SLSC-based techniques and the collected datasets used for training lack diversity in terms of skin types, ethnicities, and geographical locations, which can affect the generalizability of the models. It needs to be solved by collecting the skin lesion images from a benchmark or standard database to get a larger dataset. It can efficiently resolve the above-mentioned issues and be utilized to improve the generalizability of the model.
- While the existing models can achieve high accuracy, there is a lack of interpretability and explainability in their predictions, making it difficult for healthcare professionals to know the reason behind the model's decisions. So, advanced deep learning-based segmentation and classification approaches are introduced in this work with the vision transformer in the segmentation phase for analyzing each feature and MHA in classification makes, which the model to be more interpretable.
- Existing models often struggle with accurately segmenting uncommon skin lesions due to limited training data, posing a challenge in diagnosing and treating these conditions. Therefore, this has to be solved by the incorporation of diverse deep learning models because the adaptive layers and visual transformer network are utilized to offer highly accurate and precise outcomes for this analysis and significant advantages for skin lesion segmentation. It helps with early diagnosis and provides better treatment for skin lesion diseases.
- The traditional SLSC models require significant computational resources and time for processing, limiting their practical application in real-time scenarios such as telemedicine or mobile health applications. Thus, the optimization algorithm is suggested for better performance where the tuning of variables for the deep learning strategies is utilized to resolve the computational issues and it also makes suitable for real-time applications.
- The performance of conventional classification models can be affected by the presence of noise or low-quality images, requiring further research on techniques to handle such scenarios. So, an adaptive concept is offered using deep learning models for the classification process. A dilated layer is connected with the DenseNet to provide a good classification performance and also an advanced feature of the median filter process is offered for better skin lesion classification that effectively removes noise from dermoscopic images.

Author [citation]	Techniques	Superiorities	Downsides
Xie et al.[24]	MB-DCNN	This model has localization capability, which is used to attain more accurate lesion segmentation. It has determined the usefulness of common bootstrapping for image classification and segmentation	Simplifying the training process and enhancing the discriminatory power is limited in this model
Navarro et al.[31]	Computer-aided diagnosis	This technique permits reliable measurements for the evaluation of features. It can execute the skin lesion process under various confined conditions	It is cost-effective and computational time is high
Riaz et al.[29]	LBP	This method has exploited the advantages through various segmentation categories like edge-based, region-based, and thresholding. It has also resolved the optimization issues	But, the context-aware information presented in this technique is not explained
Bian et al.[25]	Transfer learning	It necessitates less time for training the information presented in the input image It improves the efficacy of the framework effectively	However, it has lower scalability which reduces the system efficacy rate
Ramadan and Aly[30]	CU-Net models	This model aids in effectively increasing the efficacy of the semantic segmentation deep technique. It is robust when assimilated over other classical models	However, it has faced complexity issues while performing
Sikkandar et al.[32]	ANFC	This method has been used to design an efficient segmentation as well as a classification framework for better validation. It offers an enriched result rate than the classical approaches	Utilization of classifiers for improving performance is limited
Akram et al.[26]	FCM	This model can choose the better solution depending on the boundary connections. It minimized the dimensionality issues as well as showed better outcomes	It necessitates more time to process and prepare the system
Thapar et al.[27]	GOA	This model has provided an optimal global solution and resolved the optimization issues It gives higher accuracy values for detecting skin lesions	There is a need for datasets to address real-world applications

Table 1. Advancements and limitations of conventional SLSC framework.

Proposed methodology
Implemented skin lesion segmentation and classification model

The human skin consists of three different skin tissues such as hypodermis, dermis, and epidermis. The epidermises hold melanocyte, which generates more melanin. The irregular growth of melanocytes is subjected to melanoma which is the deadliest skin cancer type. The methodical representations of the developed SLSC framework are presented in Fig. 1.

An innovative SLSC method is implemented by deep learning techniques to enhance the SLSC rate for effective diagnosis between persons in the starting stage. Different images for the validation are taken from benchmark sources. Here, median filtering and CLAHE approaches are suggested for employing the preprocessing of images. Next, it is forwarded as the input to the AL-VTransUNet-based segmentation model, and the constraints of TransUNet like batch size, n-layer, learning rate, and dropout are optimized using developed IRP-GSO to attain a better segmentation rate by maximizing the dice coefficient. Next, the segmented images are offered to the DD-MHA-based skin lesion classification model for categorizing the skin lesions effectively. Here, the parameters of dilated DenseNet are optimized to enhance the accuracy and precision rate using IRP-GSO. Thus, the recommended skin lesion framework achieved a better skin lesion classification rate than the existing model through multiple experimental observations.

Novelty of proposed model: The AL-VTransunet model is developed to enhance the segmentation accuracy and help to achieve better skin lesion results. Here, the basic UNet model utilized a limited amount of labeled images and global location for the observation to offer a better segmentation rate. The adaptive layers and visual transformer networks are utilized to offer highly accurate and precise outcomes for the analysis and also reduce the interpretability and scalability issues. It enhances the segmentation accuracy and provides better treatment for skin lesion diseases. In addition, the DD-MHA is developed to perform the skin lesion classification. A dilated layer is connected with the DenseNet to provide a better efficacy rate. Moreover, the Multi-Head Attention (MHA) layer is connected with the dilated DenseNet. Moreover, the parameters of epochs, optimizer, step per epoch, and activation function of DenseNet are tuned by a newly developed IRP-GSO algorithm that efficiently

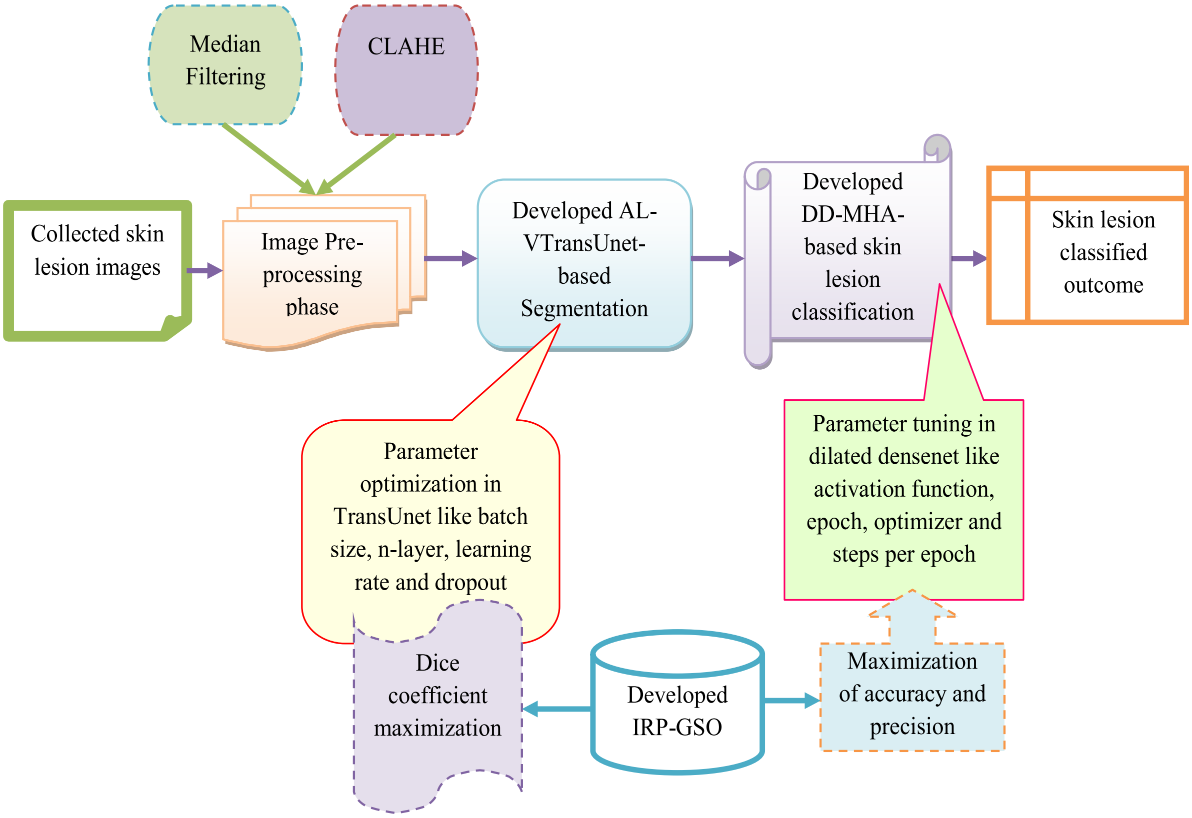


Fig. 1. Architectural design of the developed SLSC framework.

enhances the classification performance. The dilated DenseNet architecture and dense connections between layers are effectively used, which helps in feature reuse and gradient flow.

Dataset collection for experiments

Dataset 1(Harvard Dataverse)[43]: The dataset 1 is taken from "https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:https://doi.org/10.7910/DVN/DBW86T: Access date: 2023-03-18". This dataset contains mult i-resource dermoscopic images of normally pigmented skin lesions. Here, dermoscopic images are attained from diverse people, and they are preserved in multiple modalities. The dataset holds nearly 10,015 dermatoscopic image samples for analysis. Skin lesion types in the dataset 1 are benign keratosis-like lesions and dermatofibroma melanoma.

Dataset 2(PH2Dataset)[44]: The dataset 2 is collected from https://www.dropbox.com/s/k88qukc20ljnbuo/PH2Dataset.rar?file_subpath=%2FPH2Dataset: Access date: 2023-03-17. This dataset holds more dermoscopic and lesion images for observation. Here, the images are collected from different individuals, and they are commonly utilized for effective skin lesion analysis. Skin lesion types in the dataset 2 are atypical nevus, common Nevus, and melanoma.

The Benchmark datasets like Harvard Dataverse and PH2 Dataset are widely applied for skin lesion segmentation and classification by ensuring with high-quality dermoscopic images. Many medical imaging datasets does not have the ability to represent the diversity of global populations and causes potential biases that greatly affects the performance. Thus, it explores the possible biases in these datasets and mitigation strategies for enhancing the generalization ability. The potential bias from the two datasets is represented as follows.

- *Ethnicity Representation*: Datasets often have a higher representation of fair-skinned individuals, which introduces bias in segmentation models. Lesion appearance can differ across skin tones due to variations in melanin concentration, affecting lesion boundary visibility. Underrepresentation of darker skin tones may lead to lower model accuracy for these groups.
- *Potential Labeling Bias*: If the ground truth annotations in the datasets are derived predominantly from lighter-skinned patients, the model may be biased toward detecting lesions with high contrast against fair skin.

Here, the images are represented to all types of skin and ethnicities. If the model is trained predominantly on fair skin, lesion boundary detection on darker skin tones may be less accurate due to lower contrast between healthy and affected areas. Lesions generally appears on different skin tones, possibly leading to misclassifications, if the model has not been exposed to sufficient samples from diverse skin tones. The model's performance in real-world settings may decline if the test dataset has a significantly different skin tone distribution than the training dataset.

Multiple images acquired for the analysis of SLSC are presented as Im_m^{ip}, where, $m = 1, 2, ..., M$ and the acquired skin lesions are presented as M. Image samples achieved from the datasets are offered in Fig. 2.

Adaptive skin lesion segmentation model with adaptive layer-based visual transformer with UNet architecture

Skin image preprocessing

The skin images Im_m^{ip} attained from benchmark resources are given as the input to the image preprocessing technique. Efficient image preprocessing is achieved by median filtering and the CLAHE technique.

Median filtering[45]: The obtained skin images Im_m^{ip} are fed to median filtering as input. It is a non-linear smoothing approach that effectually diminishes the blurred edges available in the image. The major advanced feature of using a median filter for skin lesion segmentation is effectively removing noise and hair artefacts. It preserves the significant lesion edge information, which enhances the accuracy of the segmentation approach. It is crucial for efficiently recognizing the boundaries of skin lesions in images with high noise levels. Additionally, it manages the presence of color changes in the same lesion accurately. Due to this, the median filtering technique is chosen for this work. In this work, they restore the current points presented in the images with surrounding

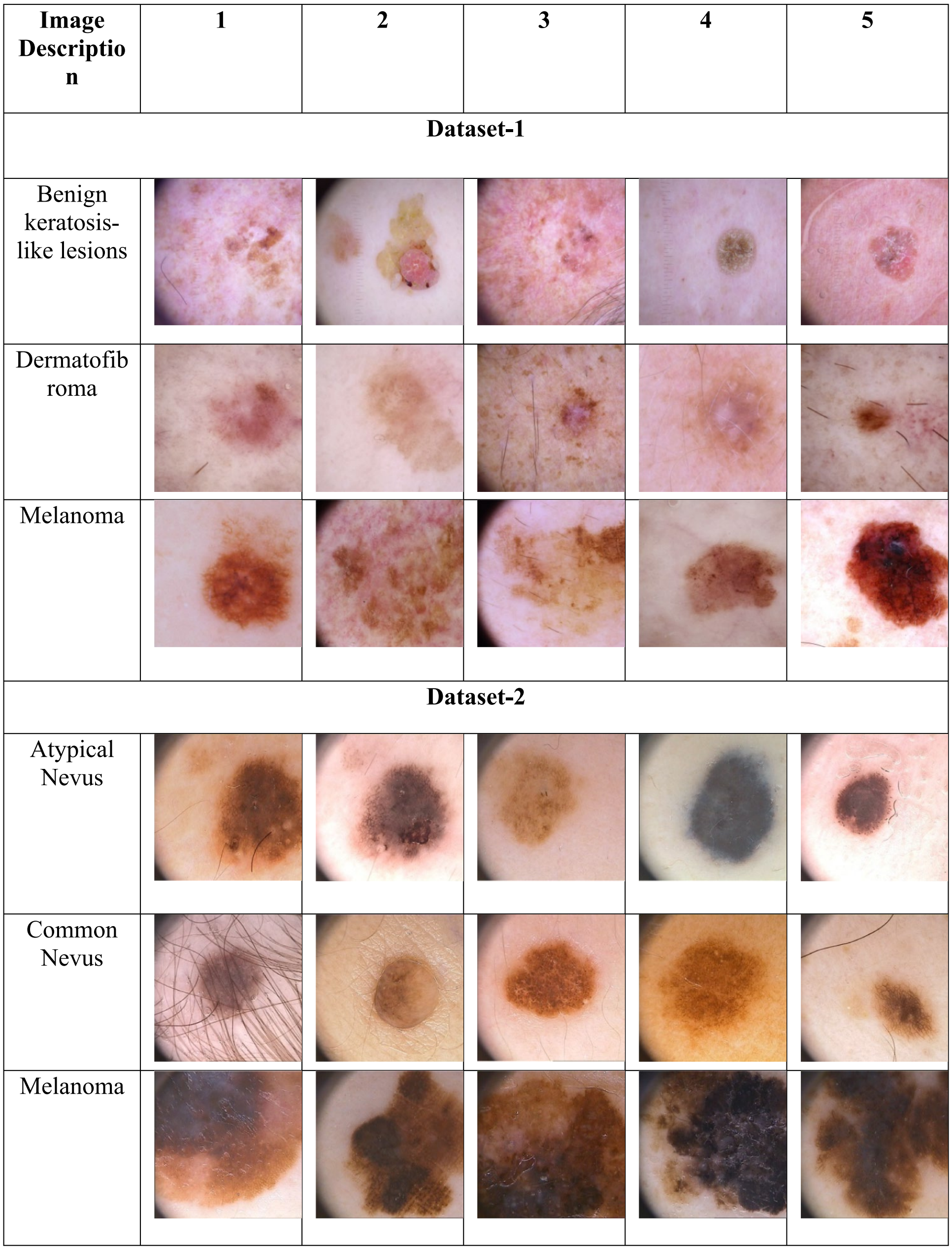


Fig. 2. Image samples from Dataset 1 and Dataset 2.

clarity in the center. Here, the individual noise spikes attained in the image cannot modify the median brightness of the surroundings. Hence, median filtering effectually ignores the impulse noise available in the input image.

Parameters determined by median filtering: The key parameter of median filtering is window size that can used to determine the neighboring pixels to calculate the median. Window size of 3 × 3 kernel (cv.medianBlur (Orig, 3)) was used in the median filtering. In general, a 3 × 3 window size plays a crucial role to eradicate the impulse noise and hair artefacts without excessive blurring. The excess usage of larger kernels (e.g., 5 × 5 or 7 × 7) smooths the lesion edges, leading to loss of critical diagnostic details. Henceforth, the outcomes of 3 × 3 preserve lesion boundaries for minimizing the noise. Images collected from the median filtering are referred to as Im_m^{Mf}, and fed as the input to the CLAHE model.

CLAHE [46]: The median filtering-based skin images Im_m^{Mf} are used as the input for effective skin lesion analysis. In skin lesion images, the CLAHE technique provides an enhanced contrast improvement rate. The tiny spaces available in the input image are represented as the circulation of Rayleigh. This method is achieved by fusing the adjacent space using a bilinear interpolation to reject the synthetically induced surroundings. Owing to this, the CLAHE approach is chosen for this work. Later, a new image is developed by combining the CLAHE model with the input image provided in Eq. (1).

$$\mathrm{Im}_m^{Mf} = \alpha PL\left(q,w\right) + \beta O\left(q,w;\sigma\right) * PL\left(q,w\right) + \mu \tag{1}$$

The Gaussian filter in the scale σ is fed as $O\left(q,w;\sigma\right)$, and the convolutional operator is presented as $*$. The term β is defined as the local average color presented in the image. Different parameters and values used in the observation are presented as $\sigma = -4$, $\alpha = 300/30$, $\mu = 4$ and $\beta = 128$. Moreover, their pattern of local averaging mapping is 50% in grayscale for μ phase.

Parameters determined by CLAHE: The parameters determined by the CLAHE are clipLimit = 2 (cv. createCLAHE (clipLimit = 2)). Considering the clip limit of 2 prevents over-amplification of noise for maximizing the local contrast of the image. Enabling higher clip limits (e.g., 3–5) results in excessive contrast enhancements, sometimes exaggerating normal skin variations. Empirical analysis on benchmark datasets (e.g., Harvard Dataverse and PH2Dataset) showed that clipLimit = 2 provided balanced contrast improvement without introducing artificial artefacts.

These preprocessing models significantly impact the quality and robustness of image segmentation by ensuring the input data is standardized, optimized, and free from noise. It allows the segmentation model to focus on identifying relevant features and produce more accurate results, especially when dealing with diverse image conditions. Good preprocessing techniques can enhance the model's capability to generalize and perform consistently across diverse situations. Ultimately, the obtained pre-processed images are indicated as $\mathrm{Im}_q^{\Pr e}$, which are provided as the input to the SLS framework.

Developed AL-VtransUNet-based skin lesion segmentation

Accurate segmentation is ensured by classifying each pixel in an image for extracting the specified regions by considering the relevant features. In this research work, the UNet[47] and Visual transformers[48] model have been adopted to enhance the segmentation performance. In addition to this, the UNet model is a popular segmentation technique that provides better accuracy and shows classification efficiency. By considering its encoder and decoder structure, the UNet model could have the ability to capture the intrinsic patterns that lead to performing well in the classification process. While training with limited or imbalanced datasets, the UNet model is prone to overfitting and degrades the system performance. Also, the vision transformer model is considered in this work to learn the long-term dependency patterns from the collected images. With the help of the attention mechanism, the vision transformer can understand the specified regions to enhance the decision-making performance to improve the model's interpretability. However, the vision transformer model is computationally expensive while training the large amount of data and leads to increased memory usage. Despite the advancement in UNet and vision transformer model, the research work leverages the AL-VTransUNet by integrating these two techniques.

Novelty of the developed model: The resultant images $\mathrm{Im}_q^{\Pr e}$ are fed as input to the skin lesion segmentation phase. In this phase, efficient segmentation is executed by the developed AL-VTransUNet. The basic UNet model utilized a limited amount of labeled images and global location for the observation to offer a better segmentation rate. But, its working procedure is slow and requires more time for training the information. To overcome the issues mentioned above, the adaptive layer and visual transformer network are connected with the UNet model. In the developed AL-VTransUNet, the TransUNet parameters like batch size, dropout, rate of learning, and n-layer are selected optimally in different ranges by IRP-GSO. Here, the parameters of TransUNet, such as batch size are modified in the series of $[4, 64]$, n-layer is tuned in the limit of $[5, 15]$, the learning rate is selected in the bound of $[0.01, 0.99]$, and the dropout is optimally selected in the range of $[0.01, 0.05]$ to offer better segmentation rate by increasing the dice coefficient rate using IRP-GSO. The fitness function of AL-VTransUNet-based segmentation is represented in Eq. (2).

$$ft_1 = \underset{\left\{bs_m^{tu}, nl_z^{tu}, lr_x^{tu}, do_c^{tu}\right\}}{Arg\max} \left(\frac{1}{dc}\right) \tag{2}$$

Here, the terms do_c^{tu} indicate the dropout, lr_x^{tu} denote the learning rate, nl_z^{tu} represent the n-layer, and bs_m^{tu} are the batch size of TransUNet. The dice coefficient is represented as dc, and it is referred to as the intersection space among the two positions presented in the segmentation process that is provided in Eq. (3).

$$dc = \frac{2 * Tu}{(2 * Tu + Tv + Tw)} \tag{3}$$

Here, the true positive is presented as Tv, the false positive is termed as Tw and the true negative is given as Tu in fitness function. The results obtained from the segmentation phase Im_p^{Seg} are given as the input to the skin lesion classification model.

Advantages of AL-VTransUNet-based sink lesion segmentation: The developed AL-VTransUNet model is utilized to enhance the segmentation performance and helps to attain better skin lesion detection outputs. Here, the adaptive layers and visual transformer network are utilized to offer highly accurate and precise outcomes, and also they easily understand the relationships among the sequential elements. It offers significant advantages for segmentation owing to its capability to effectively capture global context information through its transformer-based architecture. It enhances the accuracy of identifying complex lesion boundaries when dealing with irregular shapes, blurred borders, and variations in color within skin lesions. Owing to these, it efficiently resolves the difficulties in early-stage melanoma diagnosis due to color variations within the same lesion. Based on these advanced features, it is well-suited for classifying skin lesion segmentation tasks where precise boundary delineation is crucial. This segmentation enhances the accuracy of the model and gives the way to earlier recognition.

In-depth analysis of adaptive layer for different skin lesion characteristics:

In the developed AL-VTransUnet model, the adaptive layer dynamically adjusts its parameters against different skin lesion characteristics. This adaptability is highly driven by utilizing the IRP-GSO algorithm that effectively optimize the hyperparameters like batch size, layers, learning rate, and dropout rate based on lesion complexity.

- Irregular or complex lesions (e.g., melanoma): The adaptive layer in the developed model increases attention weights, and learning rate, and boundary refinement.
- Simple lesions (e.g., benign keratosis): The AL-VTransUnet model reduces redundant transformations, optimizing efficiency.
- Intermediate lesions (e.g., atypical nevus): It provides better balance among the feature extraction and smooth segmentation by modifying convolutional filter responses.

Mitigating overfitting using tuned parameters:

In this scenario, the tuned parameter does not cause overfitting, because it learns the training data effectively. In general, overfitting occurs when the model learns more number of training data that affects the generalization on the unseen data. The research adopts parameter tuning process to find the best value for enhancing model performance on the trained data. If it provides the cleaned trained data, the AL-VTransUNet model learns the generalizable patterns with the help of tuned parameters thus avoiding overfitting.

The symbolic view of AL-VTransUNet-based skin lesion segmentation is given in Fig. 3.

UNet

UNet[47] architecture is implemented by the inspiration of CNN and it consists of two major parts such as encoding and decoding. Next, a 2×2 max-pooling procedure is executed using step 2 in level four. In the minimal phase, they combine two 3×3 convolutional layers not including the pooling layer[49]. In this case, the decoder path is determined extensively on the right side. The graphic presentation of the U-Net model is given in Fig. 4.

Visual transformer network

Visual transformers[48] are widely utilized transformer models to effectively classify medical images. The major components of the visual transformer are the encoder and decoder. The attention function of a decoder in the visual transformer is given in Eq. (4).

$$Attn\left(A, S'\right) = soft\max\left(\frac{AS'^{P}}{\sqrt{h}}S'\right) \tag{4}$$

Here, the term S' denotes the local spatial descriptor of an encoder, $\sqrt{h}$ be the distance, $Attn$ is the attention, and A be the position of a global descriptor. The presentation of the visual transformer network is presented in Fig. 5.

Development of dilated densenet with multi-head attention mechanism for skin lesion classification framework with heuristic support

DenseNet

The DenseNet[50] framework is designed by fusing the patterns, and also this model is mainly used to increase the throughput rate of the network. Further, the dense layer presented in the network could not reduce the count of the constraint available in the network, and it makes the training simple by neglecting the unused feature map.

Implemented DD-MHA-based skin lesion classification

In the DD-MHA-based SLSC framework, segmented images Im_p^{Seg} are utilized to carry out effective analysis. The recommended DD-MHA is widely employed to enhance the skin lesion classification rate by modifying the constraints like epochs, optimizer, step per epoch, and activation function by IRP-GSO. The developed DD-MHA is a powerful approach for skin lesion classification. It combines two key techniques to enhance the classification

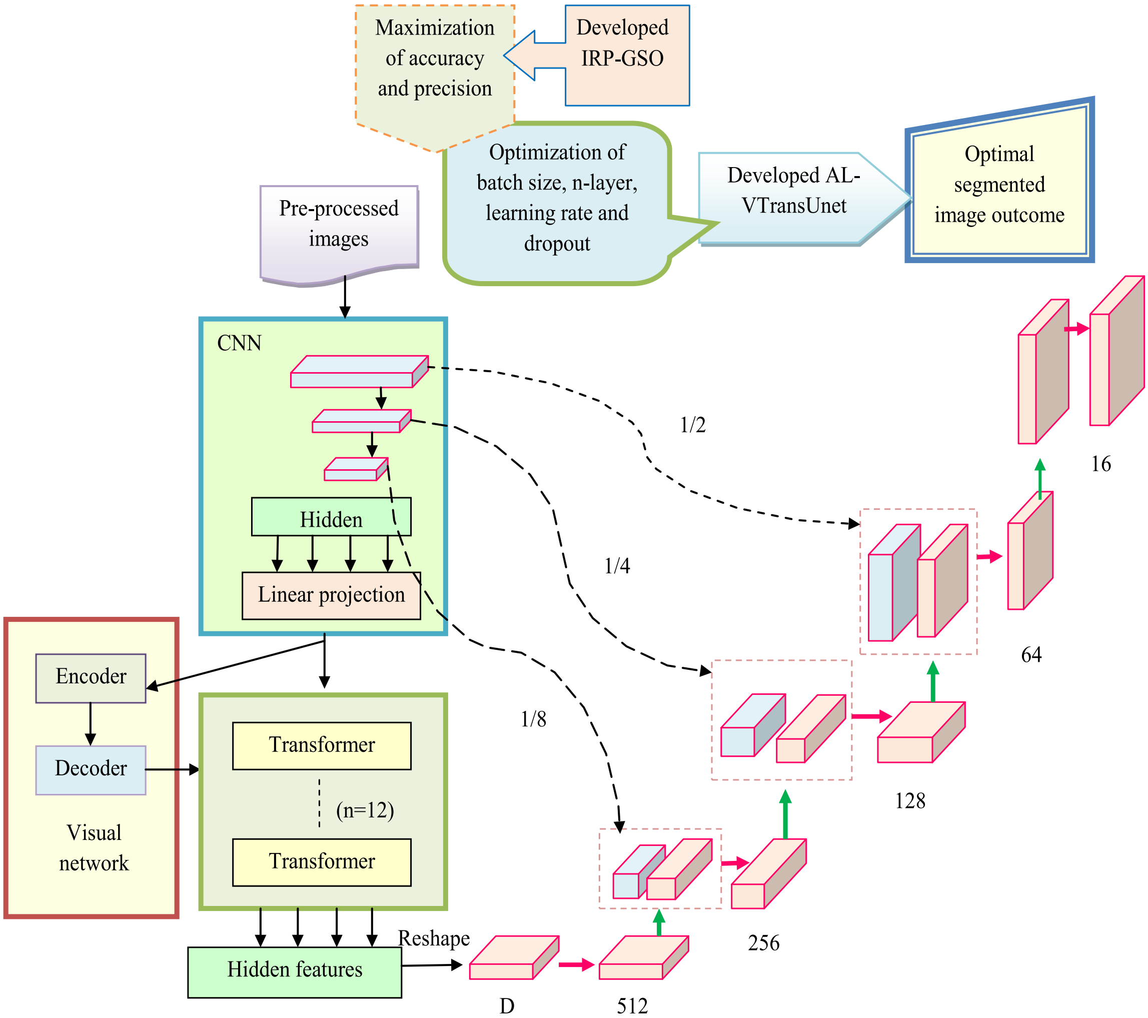


Fig. 3. Developed AL-VtransUNet-based Skin Lesion Segmentation Model.

procedure. First, the dilated DenseNet architecture is used. DenseNet is known for its dense connections between layers, which helps in feature reuse and gradient flow. By incorporating dilated convolutions, the network can capture more contextual information from the skin lesion images through the larger receptive field. This allows a better perception and representation of the image features, leading to improved classification accuracy. DenseNet has a better gradient flow rate, utilizes minimal parameters, has a better computational efficacy rate, and minimizes vanishing gradient issues. However, their implementation procedure is complex, and it needs more computational time and storage space for the analysis. To resolve these limitations obtained in the existing DenseNet, a novel deep learning model is employed in this research work. By implementing the developed model, the dilated layer has the ability to expand the receptive field without increasing the parameters to capture the multi-scale spatial features. It allows recognition of the skin lesion patterns and fine details from the images. Here, the MHA mechanism assigns different attention weights to various regions of the input images. Thus, it relatively selects the more relevant features by evaluating the multiple attention scores in parallel. In this context, the weighted combination of the attention scores is represented as output whereas, the network focus the crucial characteristics of skin lesion. Furthermore, the parameters like an epoch, optimizer, step-per-epoch and activation function are optimized with the help of IRP-GSO algorithm. It enables to adopt its weight distribution optimally to enhance the classification performance of skin lesion. Overall, the combination of dilated convolutions for spatial awareness and MHA for selective feature importance makes DD-MHA more effective for accurate and interpretable skin lesion classification.

Advantages of the Implemented DD-MHA-based Skin Lesion Classification: A dilated layer is connected with the DenseNet to provide a good performance rate in different observations. Moreover, the MHA layer is linked with the dilated DenseNet. Attention mechanisms are used in various tasks to focus on significant regions of the input. In the context of skin lesion classification, multi-head attention permits the model to selectively concentrate on different regions of the image, giving more weight to the relevant features. This helps in capturing fine-grained details and discriminative features that are crucial for accurate classification. Here, a similar group of keys, values, and queries are provided as the input, and these features are fused by the attention mechanism based on distinguishing different ranges in the sequence. Next, the projected queries are provided to the attention

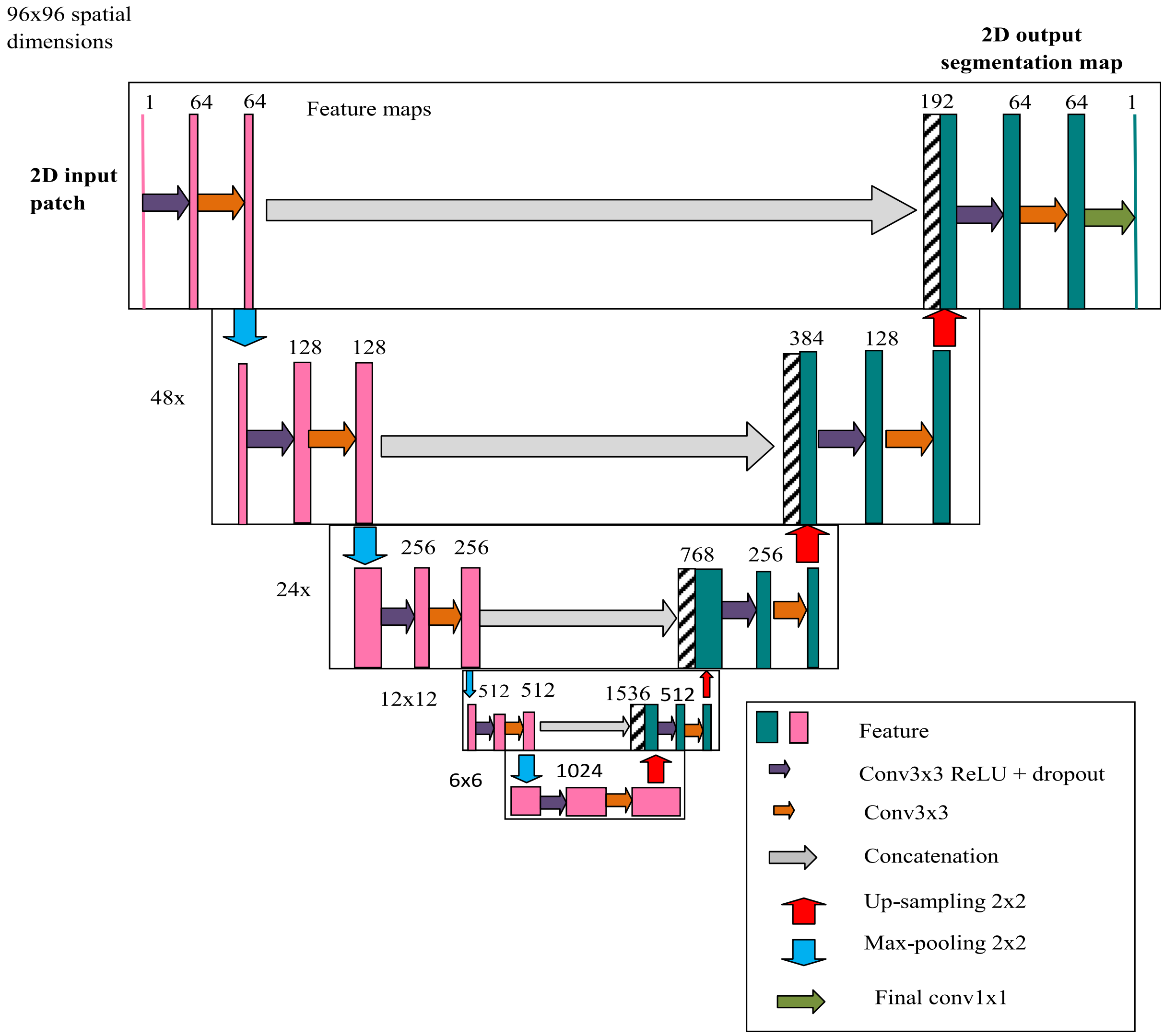


Fig. 4. Structural representation of U-Net framework.

layer, and the pooling layer to perform classification. Finally, the outcomes attained from attention and pooling layers are concatenated, and they are transformed with other learned linear projections to generate the outcome.

Objective function: In the recommended DD-MHA-based skin lesion classification model, the parameters of a dilated network, like an epoch, optimizer, activation function, and steps per count, are tuned optimally by IRP-GSO. The DenseNet parameter like an epoch is modified in the bound of [50, 100]. Optimizers are tuned effectively to select the optimizers like Rmsprop, Adam, SRD, Adagrad, and Adadelta in the limit of [0, 4], activation functions like linear, ReLU, Sigmoid, Tanh, and Leaky ReLU are optimized in the bound of [0, 4] and the steps per epoch are selected in the range of [5, 50] by IRP-GSO to increase the precision and accuracy rate.

The initial range of values for the parameters in developed DD-MHA model is selected by considering the activation function as linear, epochs as [50, 100], steps per epoch as [5, 50], and optimizer as RMSprop. Based on the loss function and data, the IRP-GSO algorithm iteratively adjusts the models parameter. Many researchers set up the initial baseline as 50 or 100. For example, if the number of epoch is selected as 50, certainly provides a good balance among the training time and enhancing the model performance. Epoch 50 means running the model through the training data with 50 times. Training too much number of epoch can leads to overfitting, so, the work adopts epoch range lies in [50, 100] to get the desired outcomes. RMSprop tends to adopt the learning rate of each parameter to provide stable convergence during the training process. In regression task, the linear activation function is often preferred for the output layers. In order to speed up the training process, the step per epoch [5, 50] is selected for the limited training data. By selecting this limited value, it is crucial to handle the training time and large dataset size whereas, it controls the batches to be processed during each epoch.

The recommended DD-MHA approach offers improved representation power, allowing to capture of more contextual information from skin lesion images. By incorporating dilated convolutions, the network can extract meaningful features and better understand the image content. Additionally, it selectively focuses on important regions of the image. This permits for better discrimination between different types of skin lesions, and enhances the classification accuracy. The recommended DD-MHA model leverages the strengths of both techniques,

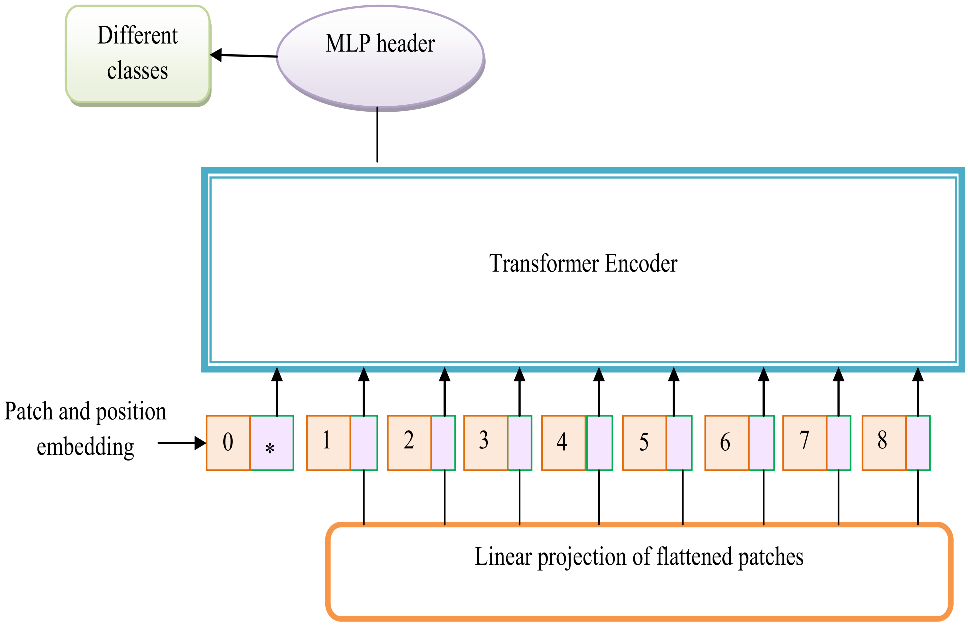


Fig. 5. Presentation of visual transformer network.

resulting in enhanced feature extraction and better contextual understanding. This ultimately leads to more accurate and reliable classification results. The fitness function of the suggested DD-MHA-based skin lesion classification model is provided in Eq. (5).

$$FT_2 = \underset{\{EP_o^{dd}, Op_i^{dd}, SE_u^{dd}, Af_h^{dd}\}}{\arg\min}\left(\frac{1}{ac + pr}\right) \quad (5)$$

Here, the terms EP_o^{dd} indicate the DenseNet epoch count, Op_i^{dd} denote the optimizer, SE_u^{dd} represent the steps per epoch, and Af_h^{dd} refer to the activation function. Precision pr is the combination of precise characteristics and the detected features, presented in Eq. (6).

$$pr = \frac{Tu}{Tv + Tx} \quad (6)$$

Accuracy ac is computed using Eq. (7).

$$ac = \frac{(Tu + Tv)}{(Tu + Tv + Tw + Tx)} \quad (7)$$

Here, the false positive is represented as Tx in the above equations. The architectural view of the recommended DD-MHA-based skin lesion classification framework is presented in Fig. 6.

The hyperparameters of the developed model with other compared models are presented in Table 2.

Proposed IRP-GSO

A novel enhanced optimization technique named IRP-GSO is employed to advance the efficacy rate of SLSC rate by modifying the parameters like learning rate, n-layer, batch size, dropout in TransUNet and epoch, optimizer, activation function, and steps per epoch in dilated DenseNet. GSO[51] is represented in a self-organized manner and it has a good scalability rate. Moreover, they are eligible to accomplish the multi-core processor instructions. Yet, they face different limitations that are detailed as follows. In the entire node, they face overhead issues, and they are highly expensive to implement and perform optimal solution searching. However, the latency rate presented between the nodes was unsuccessful in transmitting the evaluation in the sequence node. Hence, to resolve the complications presented in classical GSO, an innovative model is developed and it is named IRP-GSO. The main idea behind the recommended IRP-GSO is to enhance the traditional GSO by introducing random parameter variations. During the optimization process, IRP-GSO uses a set of randomly generated parameters. These random variations help to improve both exploration and exploitation capabilities.

Generating the variations of random parameters using IRP-GSO algorithm.

For enhancing the exploration and exploitation phases, the developed IRP-GSO optimization algorithm is implemented for introducing the random variations in hyperparameters like batch size, learning rate, dropout, and layers. Here, the random parameter is updated by utilizing the fitness concept whereas; it can perform with multiple times. Based on this process, the optimal solution is achieved using the developed algorithm. By having

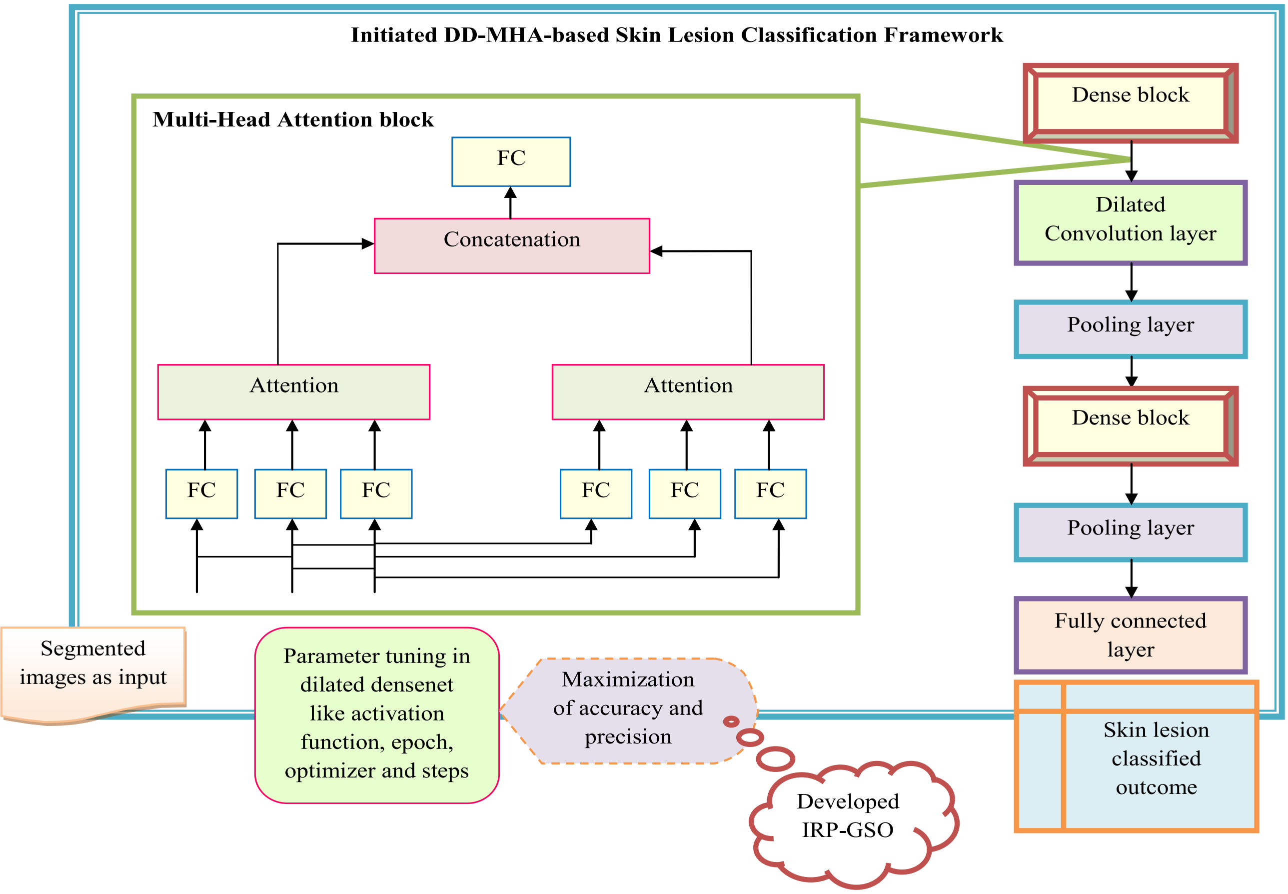


Fig. 6. Systematic representation of DD-MHA-based skin lesion classification method.

Model	Batch size	Epochs	Learning rate	Optimizer	Activation	Dropout	Layers	Steps/ epoch	Loss function
VGG16	32	50	0.001	Adam	ReLU	0.2	16 Conv + FC	5	Cross Entropy
MobileNet	32	50	0.001	Adam	ReLU	0.3	Depthwise CNN	10	Cross Entropy
XGBoost	-	-	0.1	Gradient Boosting	-	-	-	Log Loss	500 Trees
DenseNet	32	50	0.0001	Adam	ReLU	0.2	Dense Blocks	5	Cross Entropy
Developed IRP-GSO-DD-MHA	16	50–100	0.001 –0.0001	Adam, Rmsprop, Adagrad	ReLU, Sigmoid, Tanh	0.3	Dilated Densenet + MHA	5–50	Categorical

Table 2. Hyperparameters of the developed model with existing techniques.

a wider range of parameters, the IRP-GSO algorithm can search through a larger space, which increases the chances of finding better solutions. Here, the random number y are renewed with the new concept presented in Eq. (8). Here, the term y is defined as the random parameter to adjust the desirable values in a dynamic manner. Adjusting the variance dynamically allows for exploring early iterations in later stages. Also, the introduction of probabilistic step size ensures that changes occur when fitness stagnation is detected.

$$y = \left(\frac{cf - mnf}{mxf - mnf} \right) \tag{8}$$

In Eq. (8), current fitness is indicated by cf, minimum fitness is presented as mnf, and maximum fitness is termed as mxf in the recently generated concept.

Comparison of IRP-GSO with other optimization techniques and Enhancements in IRP-GSO algorithm: The downsides of existing SLSC using optimization algorithms while comparing with the proposed IRP-GSO are discussed as follows. The ACO[49] algorithm needs to focus on enlarging detection capabilities with the support of multi-modal imaging approaches and optimization algorithms[52,53]. It requires enhancing the real-world medical applicability and also needs more computational performance that enhances the model's interpretability. Need to develop an efficient CAD technique that can improve the recognition and detection of skin lesion cancer. The automatic detection of skin lesion cancer from collected images using FFOA[54,55] is a challenging process owing to the contrast among actual skin regions and skin conditions. The M-SCOA algorithm suffers from stagnation and premature convergence issues. It affects the stability of the algorithm and reduces the accuracy rate. IRP-GSO algorithm aims to define the values of parameters. The optimization of classifiers enhances the segmentation and classification accuracy. It has shown some really promising results in these areas. By incorporating random parameter variations, the developed IRP-GSO aims to provide more robust and efficient optimization solutions. It reduces the complex optimization and overfitting issues and also it enhances the cross-validation issues. The proposed IRP-GSO algorithm improved diagnosis of melanoma and early detection could result in a lower mortality rate and better treatment while comparing the above-mentioned existing approaches and the proposed IRP-GSO algorithm.

Reason for choosing IRP-GSO for segmentation and classification along with its Impacts: The IRP-GSO algorithm has been selected for fine-tuning the parameters in skin lesion segmentation and classification as it is a metaheuristic algorithm that mimics the trapping manner of glowworms. The effect of this algorithm on model performance is considerable, as optimizing the parameters could substantially enhance the segmentation reliability and classification potency, particularly while working with intricate datasets such as skin lesion images. This algorithm is preferred for its capability to effectively explore the parameter space and discover optimal configurations for both classification and segmentation models. Moreover, tuning segmentation parameters through IRP-GSO can result in better boundary recognition and the accurate outlining of skin lesions, which is vital for further experiments. This is especially advantageous when faced with unwanted or complicated images. Fine-tuning the classification parameters by utilizing IRP-GSO could improve the effectiveness of the system in successfully identifying and categorizing various skin lesion types, which is crucial for precise diagnosis and treatment strategies. Through significantly identifying the optimal parameter configurations, IRP-GSO can result in enhanced segmentation precision, better classification accuracy and minimized computational expenses where greater segmentation masks cause more precise feature extraction and assessments and fine-tuned classification parameters that contribute to superior accuracy in recognizing and categorizing various kinds of skin lesions. Also, by effectively adjusting the model configuration, the total training duration and the requirement of computational infrastructure for both classification and segmentation could be minimized. Thus, the IRP-GSO algorithm serves as an effective tool for optimizing the parameters in skin lesion classification and segmentation, resulting in enhanced system execution and effectiveness. The pseudo-code for implemented IRP-GSO is discussed in Algorithm 1.

Generate the inhabitants of the GSO
Generate GSO parameters
Update the random number y by the new concept presented in Eq. (8)
In level 1: $q_x^{(z)}, c_x^{(z)}, V_x^{(z)}$ and $g^{(z)}$ are randomly allocated
In level 2: $c^{(z)}$, $V^{(z)}$ and r are randomly assigned
For $PE \leftarrow 1$ to $PE_{\max}$
    Begin GSO: Stage 1
    For $z \leftarrow 1$ to I
        For $x \leftarrow 1$ to M
        $c_x^{(z)} \leftarrow \omega_1 c^z + j_1 y_1 \left(V_c^{(z)} -_x^{(z)}\right) + j_2 y_2 \left(g^{(z)} - q_x^{(z)}\right)$
        $q_x^{(z)} \leftarrow q_x^{(z)} + c_x^{(z)}$
          If $t\left(q_x^{(z)}\right) < t\left(V_x^{(z)}\right)$
            $V_x^{(z)} \leftarrow q_x^{(z)}$
            If $r\left(V_x^{(z)}\right) < r\left(g^{(z)}\right)$
              $g^{(z)} \leftarrow V_x^{(z)}$
              If $t\left(g^{(z)}\right) < t(g)$
                  $g \leftarrow g^{(z)}$;
              End if
            End if
          End if
        End for
    End for
    End GSO: Stage1
    Begin GSO: Stage2
    Initiate the swarm $d^{(z)} = t^{(z)}; z = 1,2,\ldots,I$;
    For $z \leftarrow 1$ to I
    $c^{(z)} \leftarrow \omega_2 c^{(z)} + j_3 y_3 \left(V^{(z)} - q^{(z)}\right) + j_4 y_4 \left(r - d^{(z)}\right)$;
    $d^{(z)} \leftarrow d^{(z)} + c^{(z)}$;
          If $t\left(q^{(z)}\right) < t\left(V^{(z)}\right)$
            $V^{(z)} \leftarrow d^{(z)}$
            If $t\left(V^{(z)}\right) < t(g)$
              $g \leftarrow V^{(z)}$;
            End if
          End if
    End for
    Stop GSO: Stage 2
End for
Return $g, t(g)$
Return optimally desirable solution

Algorithm 1: Implemented IRP-GSO.

The flowchart of recommended IRP-GSO is provided in Fig. 7.

Results and discussions
Simulation setup
An inventive skin lesion classification and segmentation technique was executed in Python. For effectual SLSC investigation, the population rate was assigned as 10, the chromosome length was taken as 4, and the iteration count was assumed as 25 to secure a good skin lesion categorization rate. In the developed SLSC framework, different existing classification models and heuristic techniques were utilized. Multiple classification techniques were compared over the developed framework including VGG16[56], MobileNet[57], XGBoost[58], and DenseNet[50], and the heuristic methods such as Tunicate Swarm Algorithm (TSA)[59], Deer Hunting Optimization Algorithm (DHOA)[20], Jaya Algorithm (JA)[60], and GSO[51] were used for performance validation. The hardware settings of the developed model are listed in Table 3.

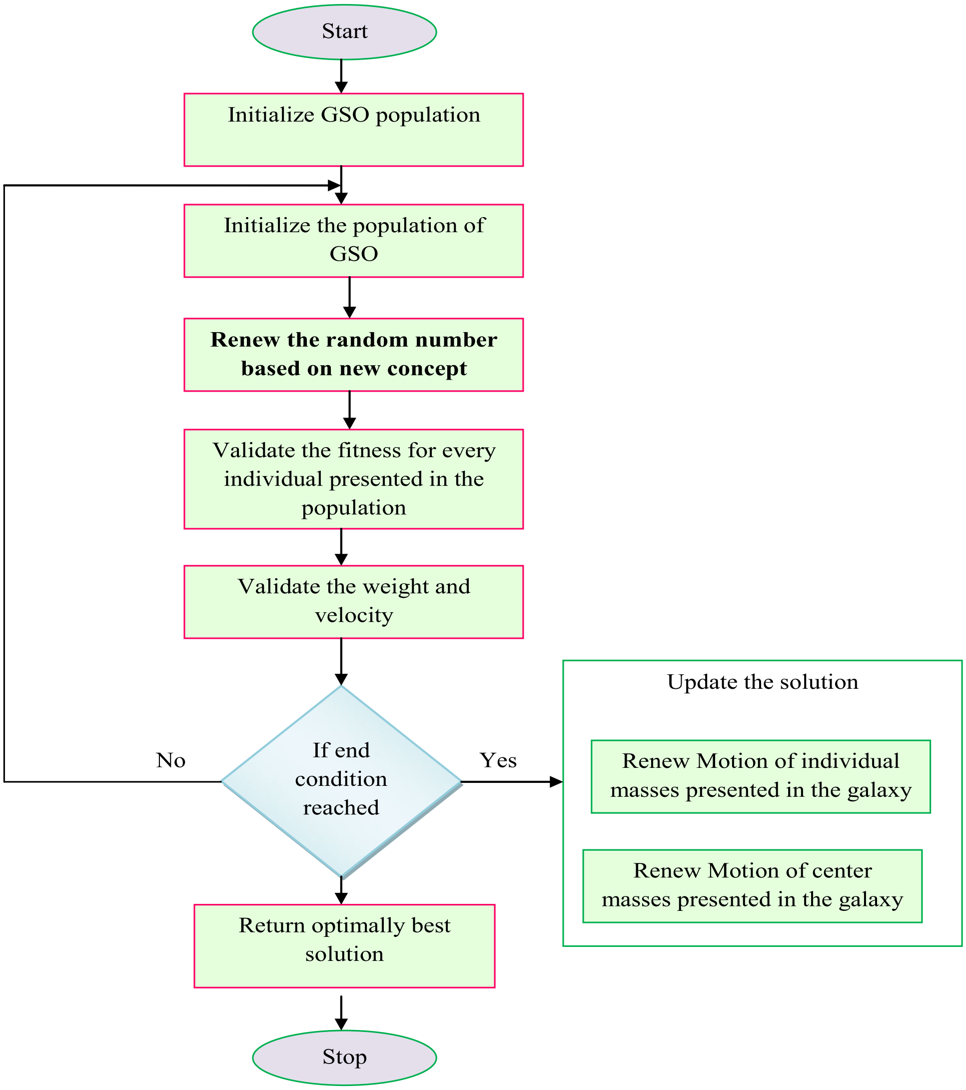


Fig. 7. Flowchart of developed IRP-GSO algorithm.

Processor	Intel(*R*) Core(TM) i3-1005G1
RAM	16.0 GB
System Type	64-bit operating system, x64-based processor
OS	Windows
Edition	Windows 11 pro
Development Environment	Python
Interpreter	Python 3.11
CPU	Intel Core i3-1005G1 @ 1.20 GHz
GPU	Intel UHD Graphics (Integrated Graphics)

Table 3. Hardware settings of the developed model.

Performance measures

Considering the complex data and models, the performance measures like False Positive Rate (FPR), Mathews Correlation Coefficient (MCC), specificity, Negative Predictive Value (NPV), False Negative Rate (FNR), F1-score and sensitivity are examined to enhance the overall performance of skin lesion classification model.

Specific pre-processing steps for calculating performance measures.

Here, the processing steps for calculating these performance measures are listed as follows.

- Removing the complex backgrounds in the inputted images could be effectively done using median filtering and CLAHE in the preprocessing phase. Ensuring a clean and quality outcome during the training process makes it easier for calculating the performance measures.
- Segmenting the images enables to provide accurate pixel level classification for boostup the processing speed while handling complex scenario in medical images.
- Tuning the model parameters based on an optimization algorithm has the tendency to enhance the model performance.
- The training process begins by considering the relevant data that is needs to be fed into the model for initiating the process.
- Data splitting: Training and testing the data tends to evaluate the model performance on the unseen data.

For attaining effective outcomes, the developed model is further validated by considering several measures for maximizing the performance in the SLSC framework. Quantitative metrics to investigate the SLSC model are elaborated below.

(a) FPR XQ is presented in Eq. (9).

$$XQ = \frac{Tw}{Tw + Tv} \tag{9}$$

While focusing the complex data, the FPR measures is calculated by considering the number of false positives by gathering the required data. From the given data, it needs to find the false positive and true negative. Here, the sum of true negative and false positive is validated. Then, the false positive gets divided by the sum of the values to get the outcomes.

(b) MCC ZW evaluates the dual recognition rate in the observation, given in Eq. (10).

$$ZW = \frac{Tu \times Tv - Tw \times Ws}{\sqrt{(Tu + Tw)\,(Tu + Tx)\,(Tv + Tw)\,(Tv + Tx)}} \tag{10}$$

MCC is the statistical tool for evaluating the performance of the model to measure the quality of binary classes. From the given Eq. (10), the numerator part proceeds with correct and penalizes incorrect prediction. Also, the denominator part specifies within the limit of -1 to + 1 to suggest the accurate outcomes.

(c) Specificity VR is represented in Eq. (11).

$$VR = \frac{Tv}{Tv + Tw} \tag{11}$$

(d) Sensitivity BT is the percentage of positive values identified and indicated in Eq. (12).

$$BT = \frac{Tu}{Tu + Tx} \tag{12}$$

While performing the specificity and sensitivity, the actual labels are required. For obtaining accurate outcomes, the pre-trained network is considered in SLSC model. In order to calculate the specificity and sensitivity, the Eqs. (11) and (12) are applied to get the relevant and reliable outcomes in SLSC model.

(e) NPV CE is presented in Eq. (13).

$$CE = \frac{Tv}{Tv + Tw} \tag{13}$$

While focusing complex data and models, the NPV is measured by comparing with the actual negative outcome and predicted negative outcome by considering the accurate and relevant data. A maximum NPV value indicates to show the model correctly identify the negative cases.

(f) FNR NY given in Eq. (14).

$$NY = \frac{Tx}{Tx + Tu} \tag{14}$$

In complex data, the FNR divides the number of false negatives as the system does not correctly identify the cancerous region whereas, the model's ability to generalize the unseen data.

(g) F1-score BR is presented in Eq. (15).

$$BR = 2 \times \frac{2Tu}{2Tu + Tw + Tx} \tag{15}$$

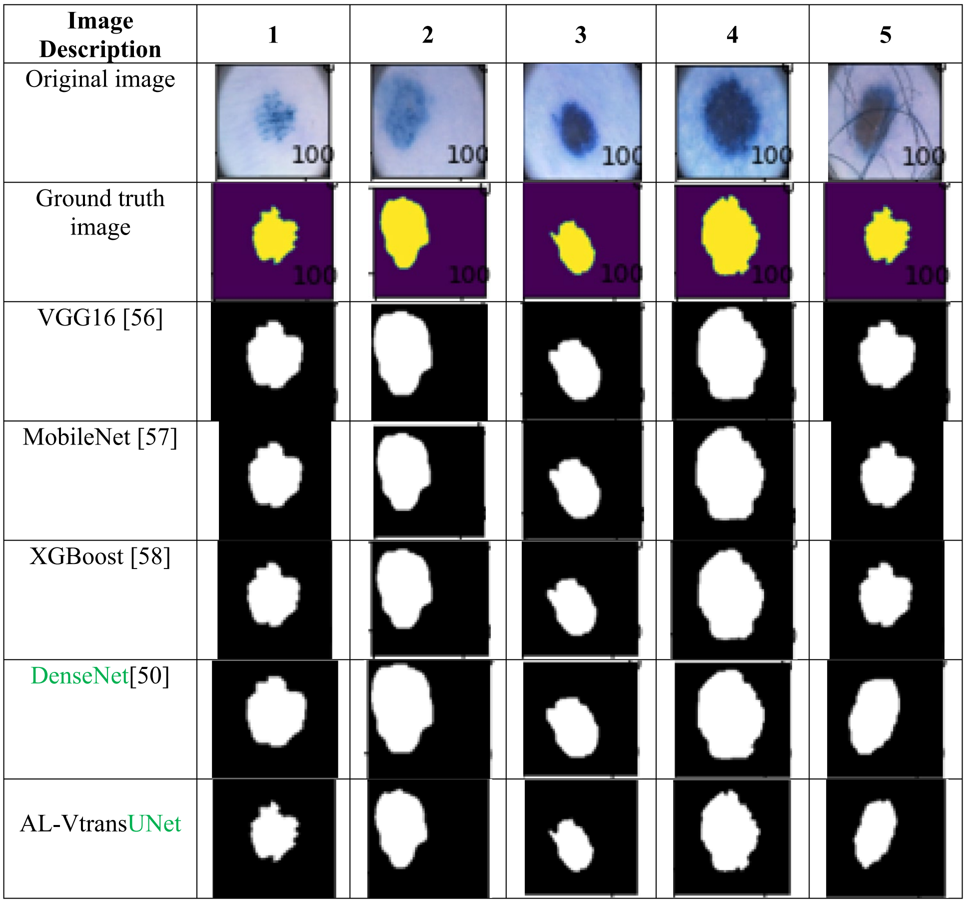


Fig. 8. Segmentation outcomes from Dataset 1.

With the help of training data, the model is performed to test/train the data. Utilizing the model prediction, the precision and recall is calculated for the process. Then, the formula of F1-score is computed to obtain accurate outcomes in the SLSC framework.

Segmented image outcomes

The AL-VtransUNet-based segmented image results achieved from dataset 1 and 2 are displayed in Figs. 8 and 9.

Efficacy analysis of implemented framework over segmentation techniques

The efficacy validation of the recommended scheme over different segmentation methods is displayed in Table 4. Jaccard analysis executed recommended SLSC framework secured good efficacy rate as 12.30%, 4.25%, 0.44%, 3.83%, 21.03%, 20.3%, and 42.59% than classical segmentation models like ResUNet, Deeplab, Region Grow, UNet, BT-GAM, Fuzzy, K-Means, and TransUNet in dataset 1. The AL-VTransUNet model has demonstrated impressive results in skin segmentation tasks, achieving high correctness and robustness. It is used in various applications, including dermatology, computer-aided diagnosis, and medical imaging.

Convergence analysis of proposed model

Convergence analysis performed on dataset 1 and 2 are displayed in Fig. 10. Convergence analysis performed on dataset 1 secured a better performance rate as 26.4%, 24.2%, 21.8%, and 19.3% better than classical techniques like DHOA-DD-MHA, TSA-DD-MHA, JA-DD-MHA and GSO-DD-MHA, correspondingly. Similarly, convergence analysis performed on dataset 2 secured better efficacy rates than the SLSC model. By accurately classifying the skin regions, the developed IRPGSO-DD-MHA model can assist in tasks such as lesion detection, boundary delineation, and analysis of skin conditions. This can have significant implications for healthcare providers, aiding in the diagnosis, treatment, and monitoring of skin diseases.

Evaluation of ROC curve on the developed model

The ROC curve of the implemented approach is represented in Fig. 11. For skin lesion classification, ROC analysis is utilized to evaluate the efficacy of a classification model. The AUC of the ROC curve provides an overall measure of the classifier's performance. Here, the ROC curve analyses are executed over truly positive values and falsely positive values in Dataset 1 and Dataset 2. Hence, the suggested IRP-GSO-DD-MHA-based

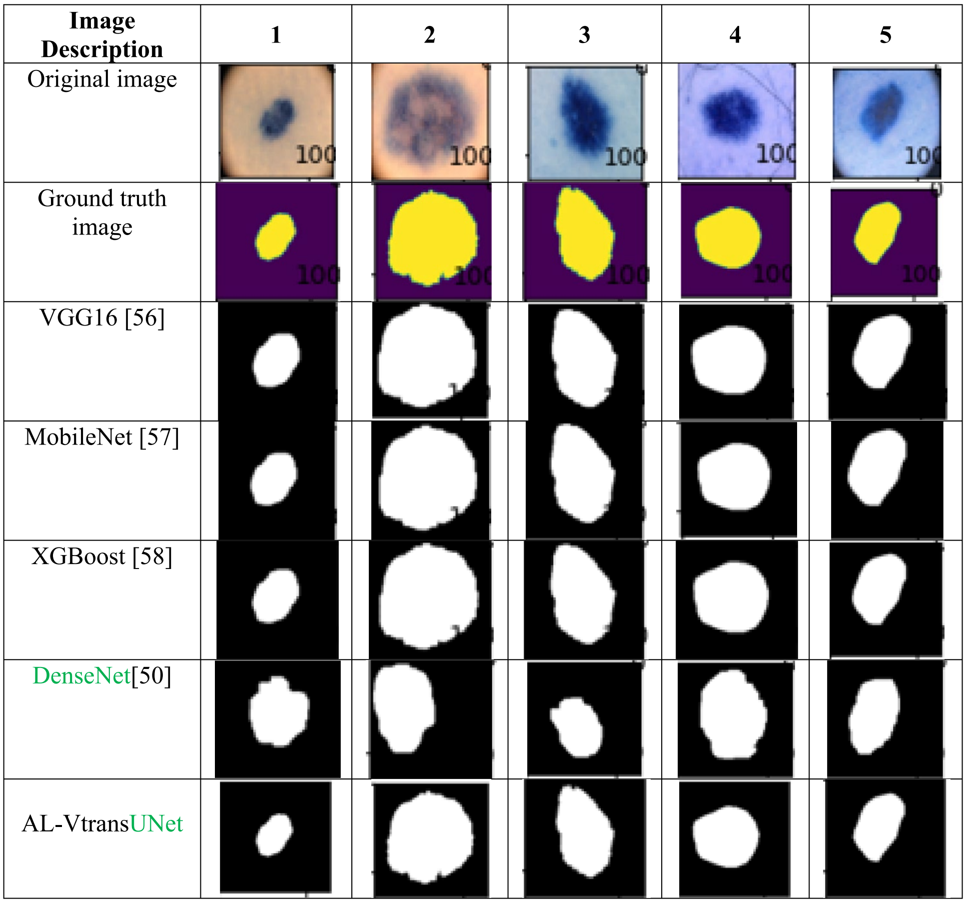


Fig. 9. Segmentation results from Dataset 2.

Metrics(%)	Fuzzy[21]	K-Means[22]	Region Grow[23]	BT-GAM[59]	UNet[47]	Deeplab[61]	ResUNet[62]	TransUNet[48]	AL-VtransUNet
Dataset 1									
Accuracy	80.27	86.22	84.44	86.04	93.13	92.78	92.84	92.56	96.49
Sensitivity	80.40	86.07	84.40	86.00	93.27	92.80	92.87	92.63	96.47
Specificity	80.20	86.30	84.47	86.07	93.07	92.77	92.83	92.56	96.49
Dice	73.09	80.64	78.34	80.42	90.05	89.55	89.64	62.41	78.55
Jaccard	57.59	67.56	64.39	67.26	81.91	81.07	81.22	45.36	64.68
Dataset 2									
Accuracy	80.33	85.91	84.62	86.04	93.27	92.89	92.89	92.67	96.50
Sensitivity	80.33	85.80	84.60	85.93	93.20	92.93	92.87	93.00	96.50
Specificity	80.33	85.97	84.63	86.10	93.30	92.87	92.90	92.50	96.50
Dice	73.14	80.24	78.58	80.41	90.22	89.70	89.70	89.42	94.84
Jaccard	57.66	67.00	64.71	67.24	82.19	81.33	81.32	80.87	90.19

Table 4. Evaluation of the SLSC technique over conventional segmentation technique in dataset 1 and dataset 2.

SLSC model scored a better performance rate. Overall, the IRPGSO-DD-MHA model offers a powerful solution for SLSC.

Performance evaluation of the developed model over dataset 1

Efficacy analysis performed on the suggested IRP-GSO-DD-MHA-based SLSC framework on dataset 1 over classical classifier and heuristic techniques are offered in Figs. 12 and 13. Accuracy analysis executed on the suggested IRP-GSO-DD-MHA-based SLSC model secured 13.25% better than VGG16, 10.5% enhanced than MobileNet, 9.3% improved than XGBoost and 6.8% increased than GSO-DD-MHA. Thus, the recommended IRP-GSO-DD-MHA-based SLSC technique secures a better performance rate than the classical models.

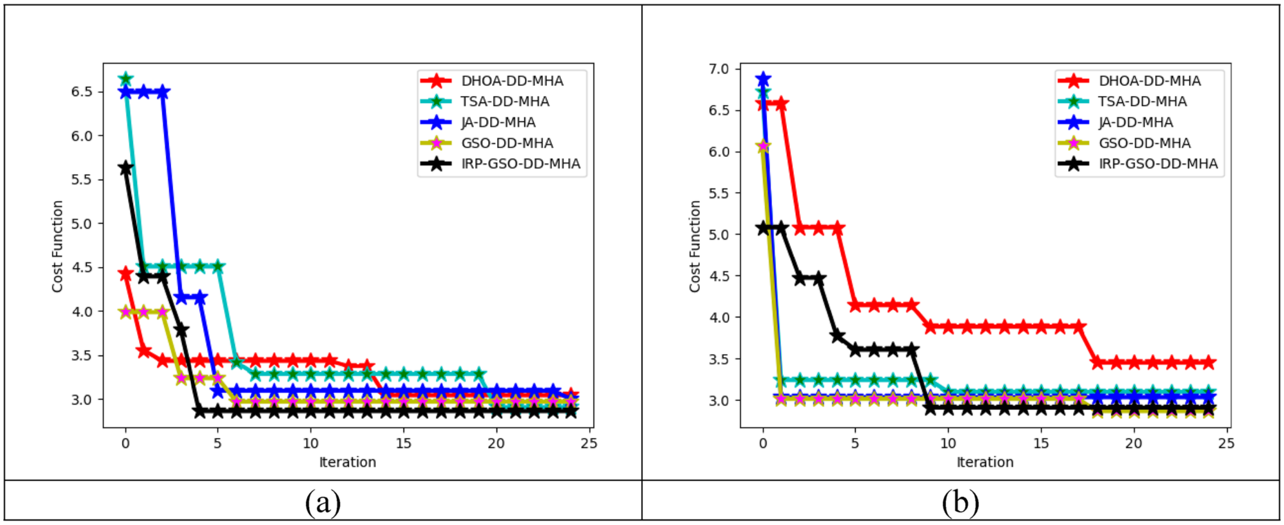


Fig. 10. Estimation on cost function for the SLSC over (a) Dataset 1 and (b) Dataset 2.

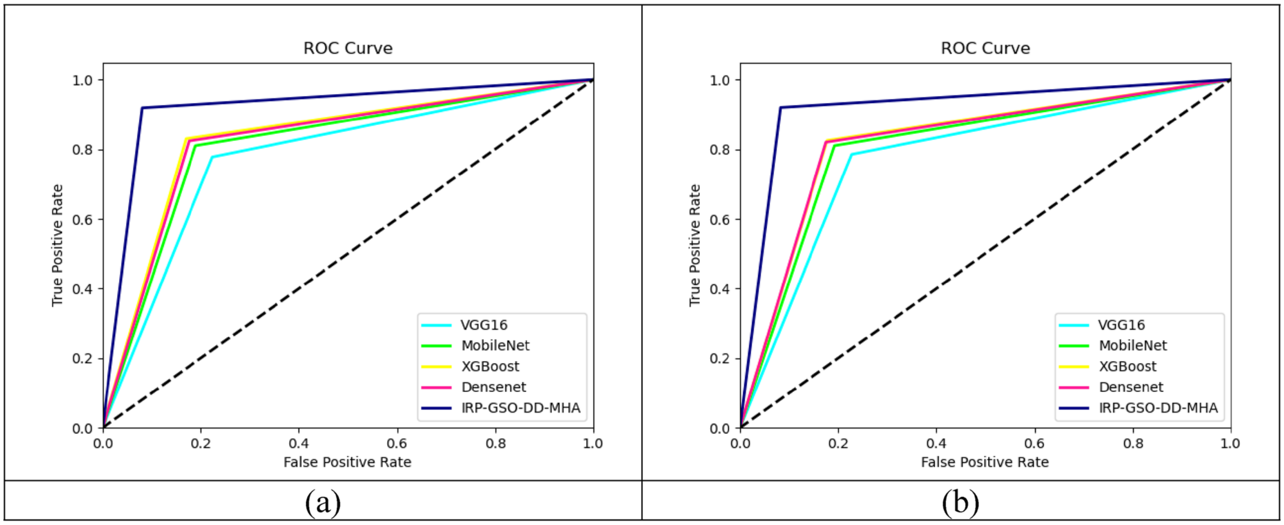


Fig. 11. Estimation on ROC Curve for the recommended SLSC methods over (a) Dataset 1 and (b) Dataset 2.

Performance validation on the recommended model over dataset 2

Efficiency analysis executed on the initiated SLSC framework over existing classification techniques and heuristic models in dataset 2 are given in Figs. 14 and 15. Precision analysis performed on IRP-GSO-DD-MHA-based skin lesion segmentation and classification framework secured better efficacy rate as 9.3%, 7.04%, 4.91% and 3.8% improved than DHOA-DD-MHA, TSA-DD-MHA, JA-DD-MHA and GSO-DD-MHA, accordingly. Similarly, multiple analyses performed on classical categorization methods secured an increased performance rate than the existing models. It shows the developed IRP-GSO-DD-MHA model allows for an increased receptive field without sacrificing spatial resolution and can handle images of varying sizes, making it adaptable to different datasets and clinical settings. Thus, the offered approach achieved a better efficacy rate.

Dataset 1-based efficacy validation on the recommended method

Dataset 1-based efficacy observation achieved on the initiated scheme over heuristic and classification techniques is given in Tables 5 and 6. Analysis of accuracy on the heuristic model in the suggested IRP-GSO-DD-MHA-based SLSC model secured 5.1% effectiveness than DHOA-DD-MHA, 3.7% improved than TSA-DD-MHA, 2.6% better than JA-DD-MHA and 1.6% superior to GSO-DD-MHA. Thus, the suggested IRP-GSO-DD-MHA system achieves state-of-the-art performance in skin lesion classification tasks, leading to timely interventions and improved patient outcomes.

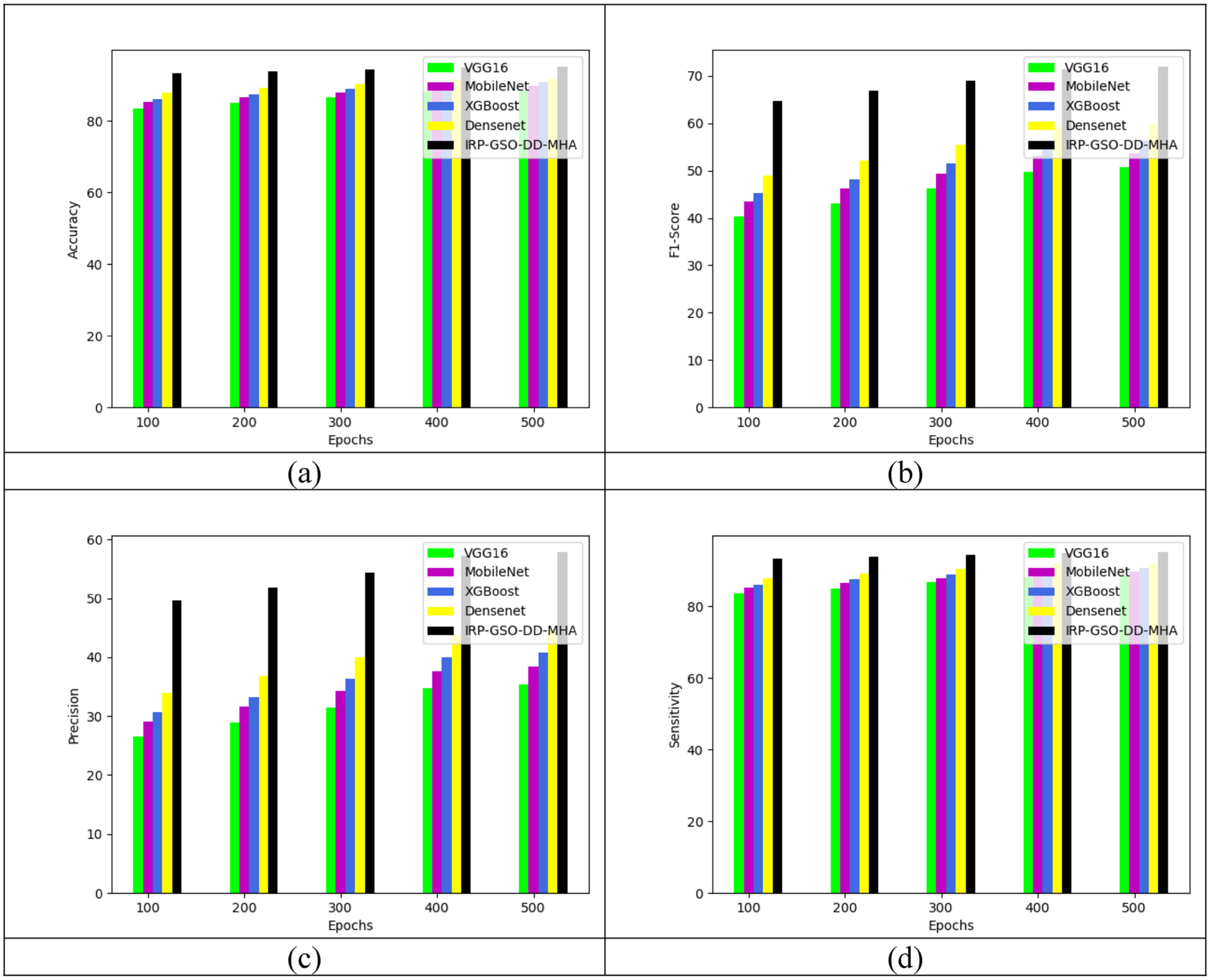


Fig. 12. Performance validation of developed SLSC model with classical models in dataset 1 over (a) Accuracy, (b) F1-Score, (c) Precision and (d) Sensitivity.

Dataset 2-based efficacy observation on suggested framework

Different efficacy analysis is executed in the proposed framework over-optimization methods and classification techniques that are presented in Tables 7 and 8. Sensitivity investigation executed on the developed SLSC framework gained a good efficacy rate of 6.14% in VGG16 and MobileNet, 4.3% more effective than MobileNet and 3.8% better than DenseNet. This proves the developed IRP-GSO-DD-MHA model reduces the need for invasive procedures by providing a non-invasive and efficient classification of skin lesions and assisting dermatologists in decision-making processes, providing them with valuable insights and support.

State-of-the-art validation of the designed model for dataset 1 and dataset 2

The efficacy validation of the suggested model is done using deep learning models for datasets 1 and 2 as given in Table 9. The efficacy validation of a SLSC model is validated using positive and negative metrics. This analysis proved that the developed segmentation technique helps to accurately segment the skin lesions across diverse kinds of conditions that provide reliable diagnostic support for dermatologists. The better accuracy rate is attained to be 95.17% from Dataset 2 for the developed AL-VTransUNet.

Computational analysis of the suggested approach

The computational analysis of the suggested approach for datasets 1 and 2 is shown in Table 10. The computational analysis of the offered approach is higher than the other baseline approaches. It is validated to show how the proposed model was completed within minimum computational time.

Accuracy validation for the suggested approach based on subtle color or texture differences

The accuracy validation of the suggested approach according to subtle color or texture differences is shown in Table 11. The subtle color or texture differences of the input images are validated based on the accuracy metrics. In this research work, the median filter and CLAHE effectively remove the noise, and hair artefacts from images

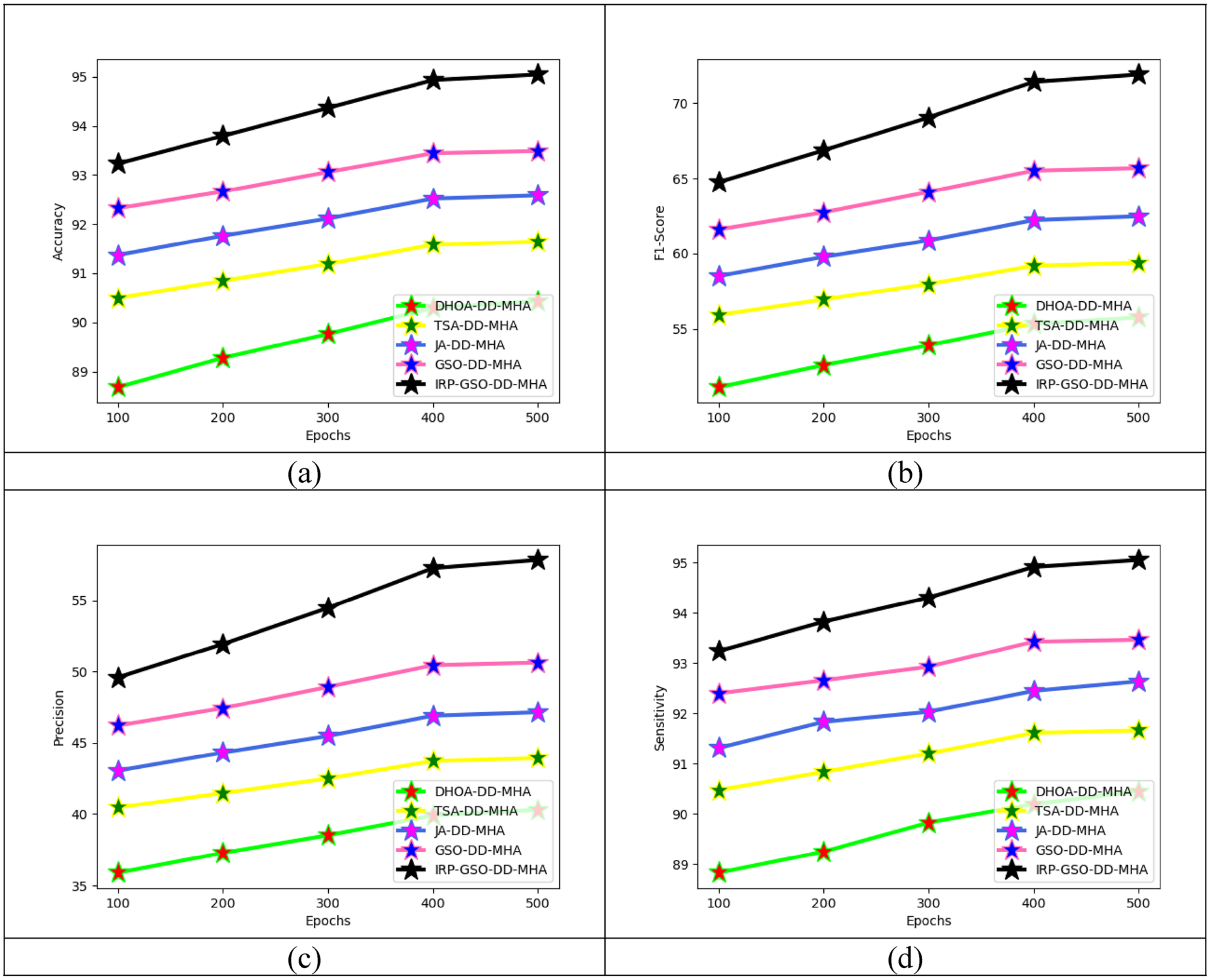


Fig. 13. Performance validation on the suggested SLSC model with heuristic technique in dataset 1 over (a) Accuracy, (b) F1-Score, (c) Precision, and (d) Sensitivity.

and hence it produces a 93.4% accuracy rate. The given table results explored that the preprocessing techniques used in this model elevated the classification performance.

Learning curves of the developed model

The learning curves of the developed model in terms of dataset 1 and dataset 2 are shown in Fig. 16. By validating with two different datasets, the training/validation-based accuracy analysis is evaluated to show the effectiveness of the model.

Complexity analysis of the developed model

Table 12 depicts the complexity analysis of the developed model by comparing with existing conventional approaches. Here, the complexity analysis is evaluated with time as well as the space complexity of the model. From the given table analysis, the time complexity of the developed model shows 41.384s and 32.208s in terms of dataset 1 and dataset 2, respectively.

The term L denotes the number of layers, feature map size is denoted as H/W. The variable C_out and C_in denotes the number of output channels and input channels. The term $T,\ N,\ K\ and\ d$ denotes the trees, number of samples, Kernel size and embedding dimension.

Statistical significance analysis of the developed model

Table 13 shows the statistical test analysis of the developed model by comparing with t-test and p-test is tested for both datasets (dataset 1 and dataset 2). It provides the crucial analysis to ensure better decision performance by quantifying the observed data. Here, the t-statistic is the crucial component to evaluate the statistically significant difference among the groups or sample mean. T-statistic is used to validate the P-value that represents the probability of the observed data if the hypothesis is true.

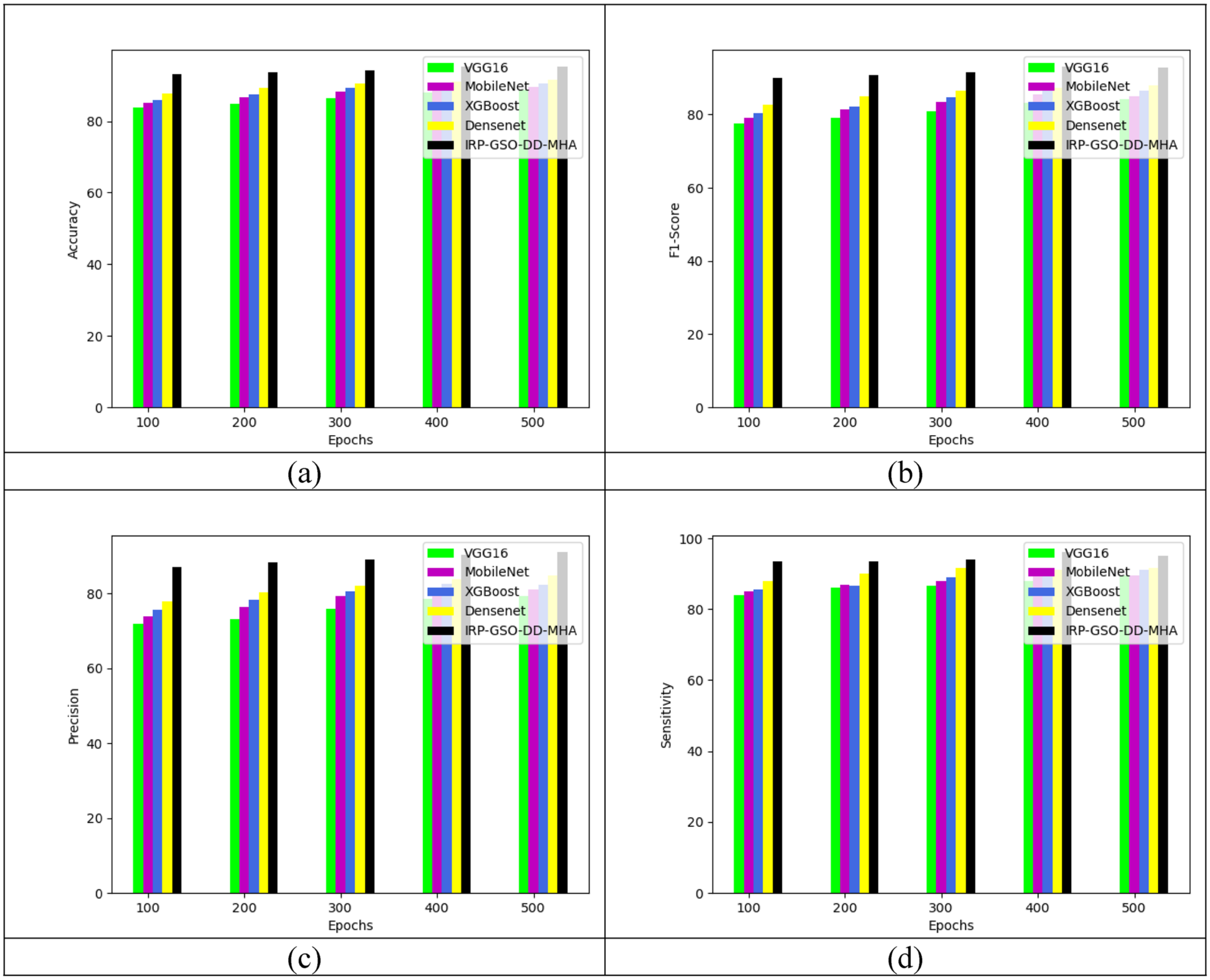


Fig. 14. Estimation on the offered framework in dataset 2 over (a) Accuracy, (b) F1-Score, (c) Precision and (d) Sensitivity.

Experimental analysis of various lesion types using the developed AL-VTransUnet model

Figure 17 represents the experimental analysis of the various types of the lesion using the developed AL-VTransUnet model and is validated in terms of dice coefficient, jaccard index and sensitivity. Here, the skin lesions like melanoma, benign keratosis, atypical nevus, dermatofibroma and common nevus are considered to validate the performance. In this experimental evaluation, the developed AL-VTransUnet model is compared against the UNet model to prove the effective outcomes for the accurate segmentation. By analyzing these metrics, the developed AL-VTransUnet model shows better qualitative outcomes to enhance the detection process.

Experimental analysis of 'with' and 'without' optimization using IRP-GSO algorithm

Table 14 shows the experimental evaluation of 'with' and 'without' optimization with the help of developed IRP-GSO algorithm. In table analysis, the standard metrics like positive as well as negative measures are validated to strengthen the performance of the model. Here, the accuracy of the model shows 95.05% in terms of with optimization. Optimizing the parameters leverages to provide better performance to handle the data imbalance issues effectively and resolve the issues of overfitting to provide faster training response in the SLSC framework. The simulation findings of the developed model provides significant advantages while reducing the error rate and enhancing model robustness.

Discussion

In the discussion section, the developed model is compared and validated against several approaches. For the precise experimental analysis, the developed model tends to evaluate the model performance, in which to prove the effectiveness in the SLSC model. While validating accuracy, the existing Fuzzy model shows 80.27% which seems to be more inaccurate than the other prior techniques. Validating the accuracy of the developed AL-VtransUNet model achieves 96.49% during the segmentation process. Timely and accurate segmentation can leads to provide accurate diagnosis of skin lesions. Ensuring better precision rate has the ability to reduce the

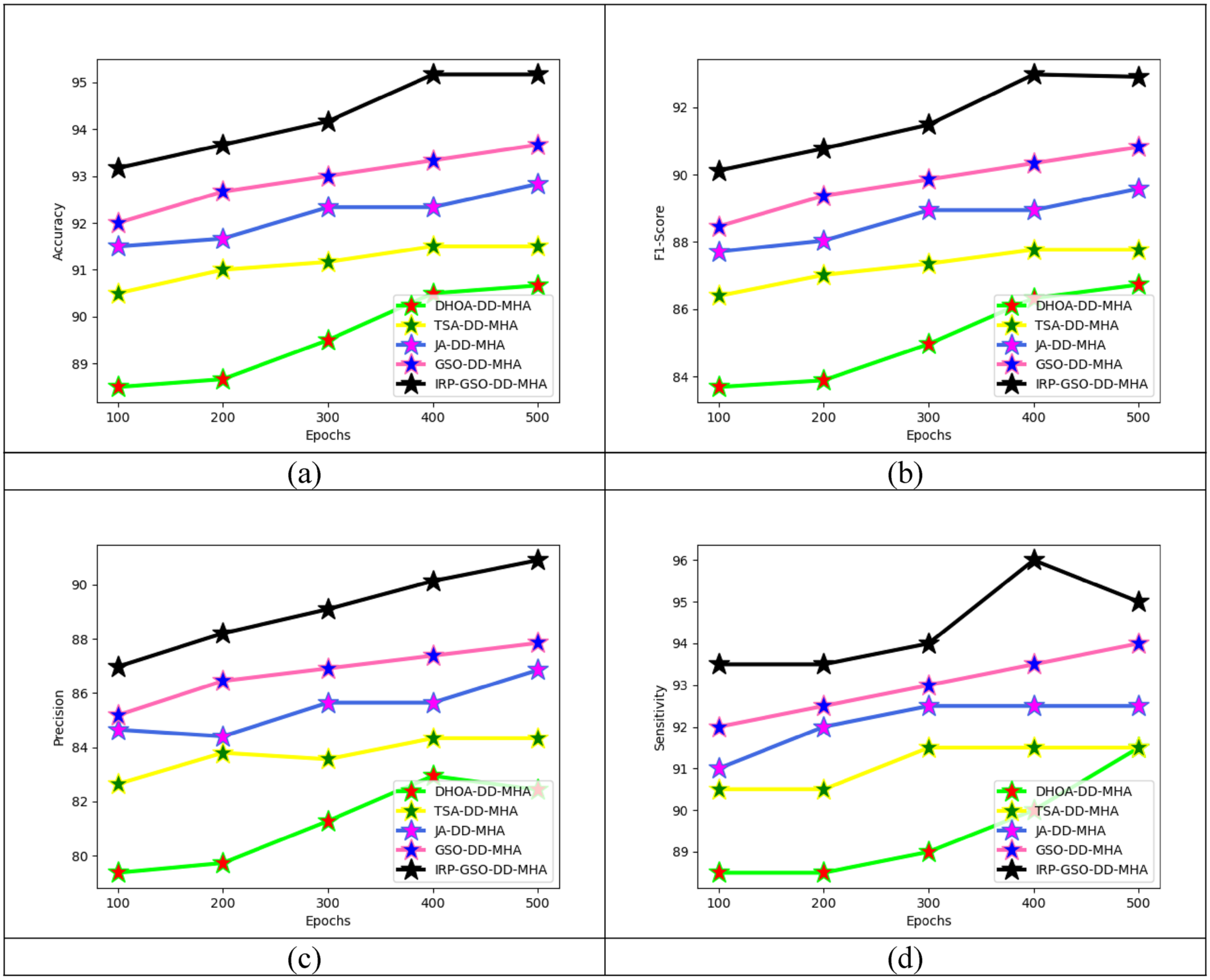


Fig. 15. Performance validation of the suggested SLSC model with heuristic technique in dataset 2 over (a) Accuracy, (b) F1-Score, (c) Precision, and (d) Sensitivity.

Measures (%)	DHOA-DD-MHA[20]	TSA-DD-MHA[59]	JA-DD-MHA[60]	GSO-DD-MHA[51]	IRP-GSO-DD-MHA
Accuracy	90.43	91.64	92.58	93.49	95.05
Sensitivity	90.45	91.66	92.64	93.47	95.06
Specificity	90.43	91.64	92.58	93.49	95.05
Precision	40.31	43.93	47.14	50.62	57.81
FPR	9.57	8.36	7.42	6.51	4.95
FNR	9.55	8.34	7.36	6.53	4.94
NPV	99.25	99.35	99.44	99.50	99.63
FDR	59.69	56.07	52.86	49.38	42.19
F1-Score	55.76	59.39	62.48	65.68	71.90
MCC	56.57	60.05	63.00	66.02	71.94

Table 5. Evaluation of the presented SLSC technique over conventional heuristic schemes in dataset 1.

false positives in the SLSC framework. Minimizing false positives can reduce irrelevant biopsies and enhance patient outcomes. Considering the existing baseline models have emerging with a huge number of false positives or false negatives, which are indicated as non-cancerous lesions mistakenly classified as cancerous. Yet, it leads to provide misdiagnosis and impact the patients quality of life. Focusing the developed model facilitates to gradually decrease the false negatives and false positives as 4.95% and 4.94%.

Comparative analysis of prior research work and the developed research work.

Measures (%)	VGG16[56]	MobileNet[57]	XGBoost[58]	DenseNet[50]	IRP-GSO-DD-MHA
Accuracy	88.49	89.70	90.58	91.84	95.05
Sensitivity	88.57	89.68	90.61	91.80	95.06
Specificity	88.48	89.70	90.58	91.84	95.05
Precision	35.45	38.34	40.73	44.55	57.81
FPR	11.52	10.30	9.42	8.16	4.95
FNR	11.43	10.3	9.39	8.20	4.94
NPV	99.09	99.18	99.27	99.37	99.63
FDR	64.55	61.66	59.27	55.45	42.19
F1-Score	50.64	53.71	56.20	59.99	71.90
MCC	51.59	54.57	56.99	60.61	71.94

Table 6. Validation of the proposed SLSC technique over classical techniques in dataset 1.

Measures (%)	DHOA-DD-MHA[20]	TSA-DD-MHA[59]	JA-DD-MHA[60]	GSO-DD-MHA[51]	IRP-GSO-DD-MHA
Accuracy	90.67	91.50	92.83	93.67	95.17
Sensitivity	91.50	91.50	92.50	94.00	95.00
Specificity	90.25	91.50	93.00	93.50	95.25
Precision	82.43	84.33	86.85	87.85	90.91
FPR	9.75	8.50	7.00	6.50	4.75
FNR	8.50	8.50	7.50	6.00	5.00
NPV	95.50	95.56	96.12	96.89	97.44
FDR	17.57%	15.67	13.15	12.15	9.09
F1-Score	86.73	87.77	89.59	90.82	92.91
MCC	79.82	81.43	84.23	86.11	89.30

Table 7. Evaluation of the implemented SLSC technique with existing heuristic scheme in dataset 2.

Measures (%)	VGG16[56]	MobileNet[57]	XGBoost[58]	DenseNet[50]	IRP-GSO-DD-MHA
Accuracy	88.67	89.50	90.50	91.67	95.17
Sensitivity	89.50	89.50	91.00	91.50	95.00
Specificity	88.25	89.50	90.25	91.75	95.25
Precision	79.20	81.00	82.35	84.72	90.91
FPR	11.75	10.50	9.75	8.25	4.75
FNR	10.50	10.50	9.00	8.50	5.00
NPV	94.39	94.46	95.25	95.57	97.44
FDR	20.80	19.00	17.65	15.28	9.09
F1-Score	84.04	85.04	86.46	87.98	92.91
MCC	75.64	77.21	79.41	81.76	89.30

Table 8. Validation in suggested SLSC scheme over traditional classification techniques in dataset 2.

Table 15 depicts the comparative analysis of the existing research works and the developed research work.

Proposed model's applicability on clinical settings

The AL-VTransUNet utilizes the strengths of both deep learning frameworks, in which the transformers are used for extracting the global contextual information and UNet for accurate segmentation, which possibly enhances the diagnosis speed and the accuracy of the detection. The combination of the UNet and the transformer offers a substantial potential for enhancing the precision and diagnosis efficiency in the healthcare environment resulting in good patient results and enhanced accessibility of treatments. The system's capability to precisely classify and segment skin lesions can help in early diagnosis and enhance patient results, which is shown in Fig. 18. According to Fig. 18(a), the traditional VGG16 model attains an accuracy of 88% which is considered to be low as compared to other models, leading to incorrect diagnosis, and prolonged or unsuitable care which reduces the public trust in this technology. However, the designed AL-VTransUNet model attains 96% accuracy which provides various advantages such as enhanced victim outcomes, minimized healthcare expenses and improved effectiveness in productivity which leads to quick and more effective medical procedures. Moreover, Fig. 18(c) shows that the existing VGG16, MobileNet, XGBoost and denseNet models demand more memory

Measures (%)	S-MobileNet[34]	LeNet[33]	UNet++[35]	DL[28]	AL-VTransUNet
Dataset 1					
Accuracy	91.21	92.12	93.04	93.97	95.05
Sensitivity	91.15	92.12	93.04	93.92	95.06
Specificity	91.22	92.12	93.04	93.97	95.05
Precision	42.57	45.50	48.85	52.66	57.81
FPR	8.78	7.88	6.96	6.03	4.95
FNR	8.85	7.88	6.96	6.08	4.94
NPV	99.31	99.39	99.47	99.54	99.63
FDR	57.43	54.50	51.15	47.34	42.19
F1-Score	58.04	60.91	64.06	67.48	71.90
MCC	58.74	61.49	64.49	67.73	71.94
Dataset 2					
Accuracy	91.00	92.83	93.17	93.83	95.17
Sensitivity	91.50	92.50	93.50	94.00	95.00
Specificity	90.75	93.00	93.00	93.75	95.25
Precision	83.18	86.85	86.98	88.26	90.91
FPR	9.25	7.00	7.00	6.25	4.75
FNR	8.50	7.50	6.50	6.00	5.00
NPV	95.53	96.12	96.62	96.90	97.44
FDR	16.82	13.15	13.02	11.74	9.09
F1-Score	87.14	89.59	90.12	91.04	92.91
MCC	80.46	84.23	85.04	86.45	89.30

Table 9. State-of-the-art validationof the suggested AL-VTransUNet scheme for dataset 1 and dataset 2.

Algorithms	DHOA-DD-MHA[20]	TSA-DD-MHA[59]	JA-DD-MHA[60]	GSO-DD-MHA[51]	IRP-GSO-DD-MHA
Dataset 1					
Time (sec)	49.4671	45.8434	42.5783	41.4577	37.2147
Dataset 2					
Time (sec)	43.7574	51.5673	45.3525	48.3463	41.5783

Table 10. Computational analysis of the suggested model.

Filtering differences	Accuracy
Original image	91.7%
Median filter image	92.5%
CLAHE image	92.7%
Gray image	92.7%
Median filter with CLACHE	93.4%

Table 11. Accuracy Estimation of the offered approach based on subtle color or texture differences.

resources which are high computational loads, making them unsuitable for running applications on low-resource devices. Also, a system with more resource utilization could cause extended training times and higher energy demands. However, the proposed AL-VTransUNet model utilizes a very small amount of memory resources which minimizes the computational requirements and enhances the scalability of the system for managing huge datasets. As a result, it is proven that the suggested AL-VTransUNet model achieves greater accuracy and better effective performance than other conventional models.

Conclusion

An innovative SLSC framework was constructed by deep techniques for reducing human error and variability in SLSC, leading to more reliable and consistent results, and could handle complex and diverse lesion characteristics, including irregular shapes, color variations, and texture patterns. The collected images were fed to the median filtering and CLAHE techniques to execute effective preprocessing. The resultant images were fed to the AL-VTransUNet-based segmentation phase, where the parameters of TransUNet were modified by the developed IRP-GSO to increase the dice coefficient. Next, the segmented images were offered to the DD-MHA-based

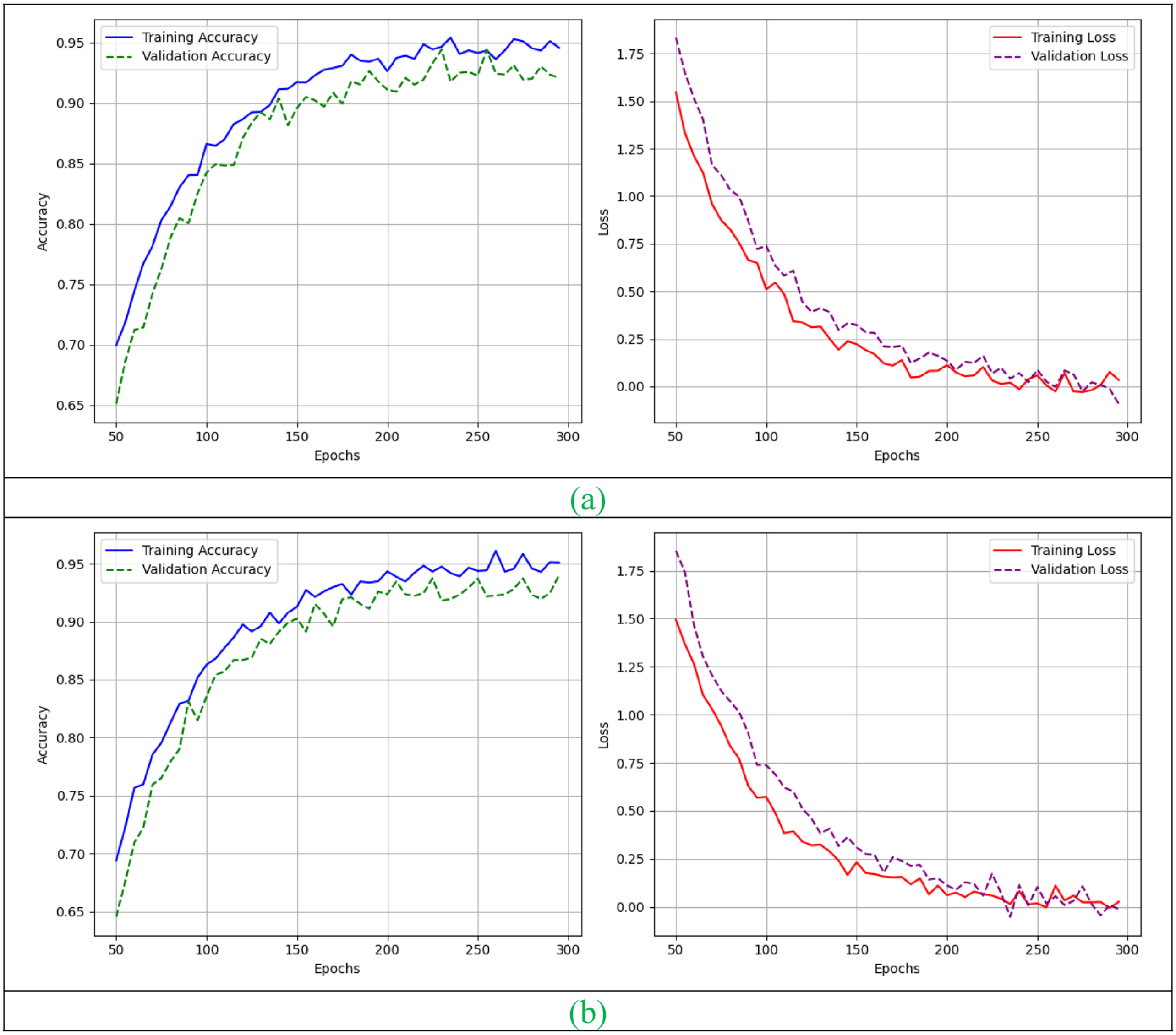


Fig. 16. Learning curves of the developed model in terms of (a) Dataset 1 and (b) Dataset 2.

Model	Time complexity (Secs)		Space complexity
	Dataset1	Dataset2	
VGG16[56]	45.837	39.838	$O(L \times H \times W \times C\_out)$
MobileNet[57]	47.368	36.992	$O(L \times H \times W)$
XGBoost[58]	44.738	34.837	$O(T \times N \times \log\ N)$
DenseNet[50]	47.993	38.009	$O(L \times C\_in) \times C\_out \times K^2$
Developed IRP-GSO-DD-MHA	41.384	32.208	$O(L \times K^2 \times H \times W) + O(n^2 \times d)$

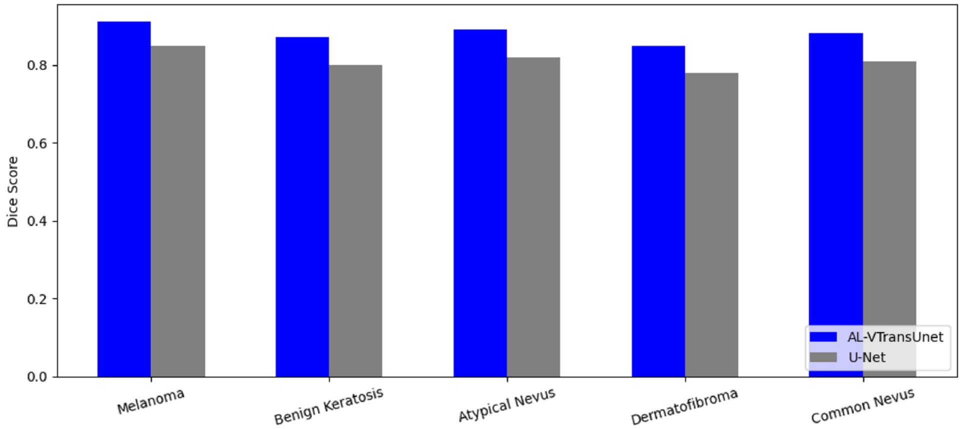
Table 12. Complexity analysis of the developed model.

classification model. Here, the parameters of dilated DenseNet were optimized by the developed IRP-GSO to enhance the precision and accuracy. Finally, the recommended SLSC framework offered an effective skin lesion-classified outcome than the classical techniques. Accuracy analysis performed on the suggested IRP-GSO-DD-MHA-based SLSC model secured 13.25% better than VGG16, 10.5% enhanced than MobileNet, 9.3% improved XGBoost and 6.8% increased than GSO-DD-MHA. Similarly, sensitivity validation employed on the developed SLSC model achieved a good efficiency rate of 6.14% in VGG16 and MobileNet, 4.3% more effective than

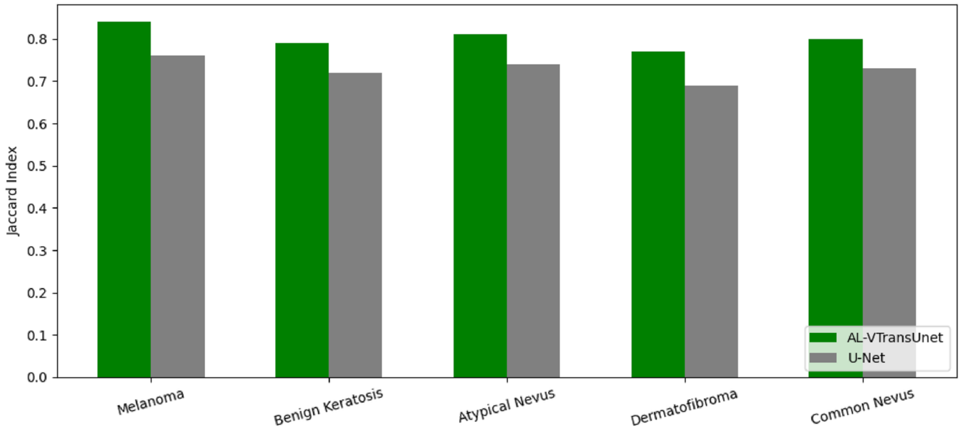
Method vs. IRP-GSO-DD-MHA	T-Statistic	*P*-Value	Significance
Dataset 1			
DHOA vs. IRP-GSO-DD-MHA	4.1234	0.0125	Yes
TSA vs. IRP-GSO-DD-MHA	4.5678	0.0097	Yes
JA vs. IRP-GSO-DD-MHA	5.0231	0.0075	Yes
GSO vs. IRP-GSO-DD-MHA	4.9823	0.0089	Yes
Dataset 2			
DHOA vs. IRP-GSO-DD-MHA	4.3456	0.0113	Yes
TSA vs. IRP-GSO-DD-MHA	4.7892	0.0081	Yes
JA vs. IRP-GSO-DD-MHA	5.1123	0.0062	Yes
GSO vs. IRP-GSO-DD-MHA	4.9234	0.0094	Yes

Table 13. Statistical test analysis of the developed model.

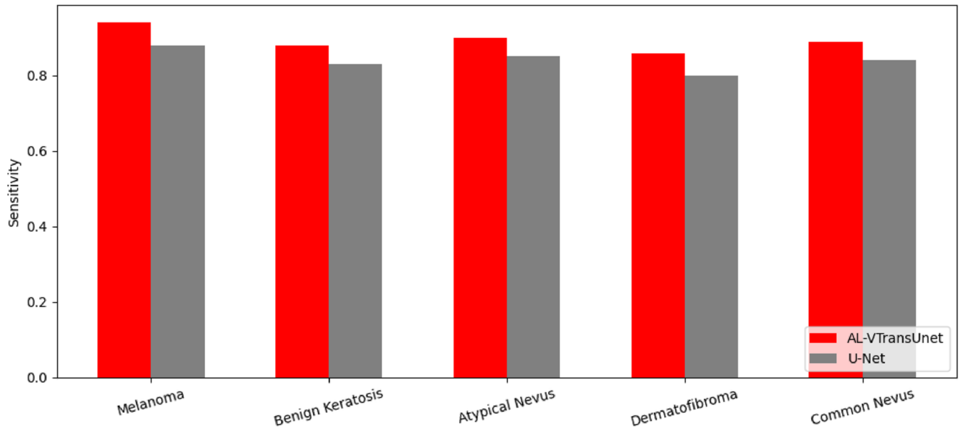
MobileNet, and 3.8% better than DenseNet. Thus, the suggested RP-GSO-DD-MHA-based SLSC model gained a better performance rate than baseline techniques in multiple observations. Finally, the experimental findings revealed the developed model could be used for automated screening of skin lesions, enabling early identification of potentially malignant cases, and had the potential to complement the expertise of dermatologists, facilitating more accurate and efficient skin lesion classification. The future scope of the developed model could involve incorporating multiple modalities of data, like dermatoscopic images, patient history, genetic information, and histopathological data. Combining these different sources of information could potentially improve the reliability and accuracy of skin lesion segmentation and classification models. It holds great promise for advancing the field of skin lesion segmentation and classification, ultimately leading to more effective diagnosis and treatment of skin conditions.



(a)



(b)



(c)

Fig. 17. In-depth analysis of various types of lesions using AL-VTransUnet model in terms of (a) Dice score, (b) Jaccard Index and (c) Sensitivity.

Measures	DD-MHA (Without optimization)	IRP-GSO-DD-MHA (With optimization)
Dataset 1		
Accuracy	93.96	95.05
Sensitivity	93.93	95.06
Specificity	93.96	95.05
Precision	52.63	57.81
FPR	6.04	4.95
FNR	6.07	4.94
NPV	99.54	99.63
vFDR	47.37	42.19
F1-Score	67.46	71.90
MCC	67.71	71.94
Dataset 2		
Accuracy	93.83	95.17
Sensitivity	94.00	95.00
Specificity	93.75	95.25
Precision	88.26	90.91
FPR	6.25	4.75
FNR	6.00	5.00
NPV	96.90	97.44
FDR	11.74	9.09
F1-Score	91.04	92.91
MCC	86.45	89.30

Table 14. Parameter optimization using the developed IRP-GSO algorithm.

Prior research work	Developed research work
In[63], the pre-trained CNN model was considered to extract the spatial information from each frame. Especially, training process in CNN model requires huge number of data and consumes more training time	In this research, the gathered images are given into the preprocessing and segmentation phase. Considering the preprocessing evaluates to generate the quality outcomes. Hence, the performance advancement in the developed model could emphasize less time to diagnose the disease
In[64], the two automated diagnosis model was implemented using an Extreme Learning Machine (ELM) based auto-diagnosis method and a hybrid method using k-means clustering and ELM for diagnosing chronic hepatitis. This ELM-based model needs to be focused on enhancing the classification accuracy of 80%	Effective classification model DD-MHA is implemented in this research to strengthen the classification performance. During the experimental evaluation, the classification accuracy of the developed model achieves 95.05% to provide accurate diagnosis in SLSC framework
In[65], the deep LSTM model is employed in this work to capture the dynamic changes of severity of disease and other clinical outcomes. It increases complexities and requires large data for validation	This research work focus on segmentation and classification in skin lesion. Analyzing the temporal attention in the deep learning model will be performed effectively to analyze the dynamic medical image sequences. For strengthening the image analysis, the developed model needs to be incorporated in the attention mechanism for getting the promising outcomes
In[66], the generative adversarial network was employed for enhancing the image segmentation. It has the ability for translating the image to perform the image segmentation. It is difficult to evaluate the quality of generated outcome	Suggesting the preprocessing and segmentation in this research work enables better quality and clear outcomes for maximizing the performance in segmentation. Thus, it leads to maximize the diagnosis treatment suggested by the medical professionals

Table 15. Comparative analysis of prior research works and developed research work.

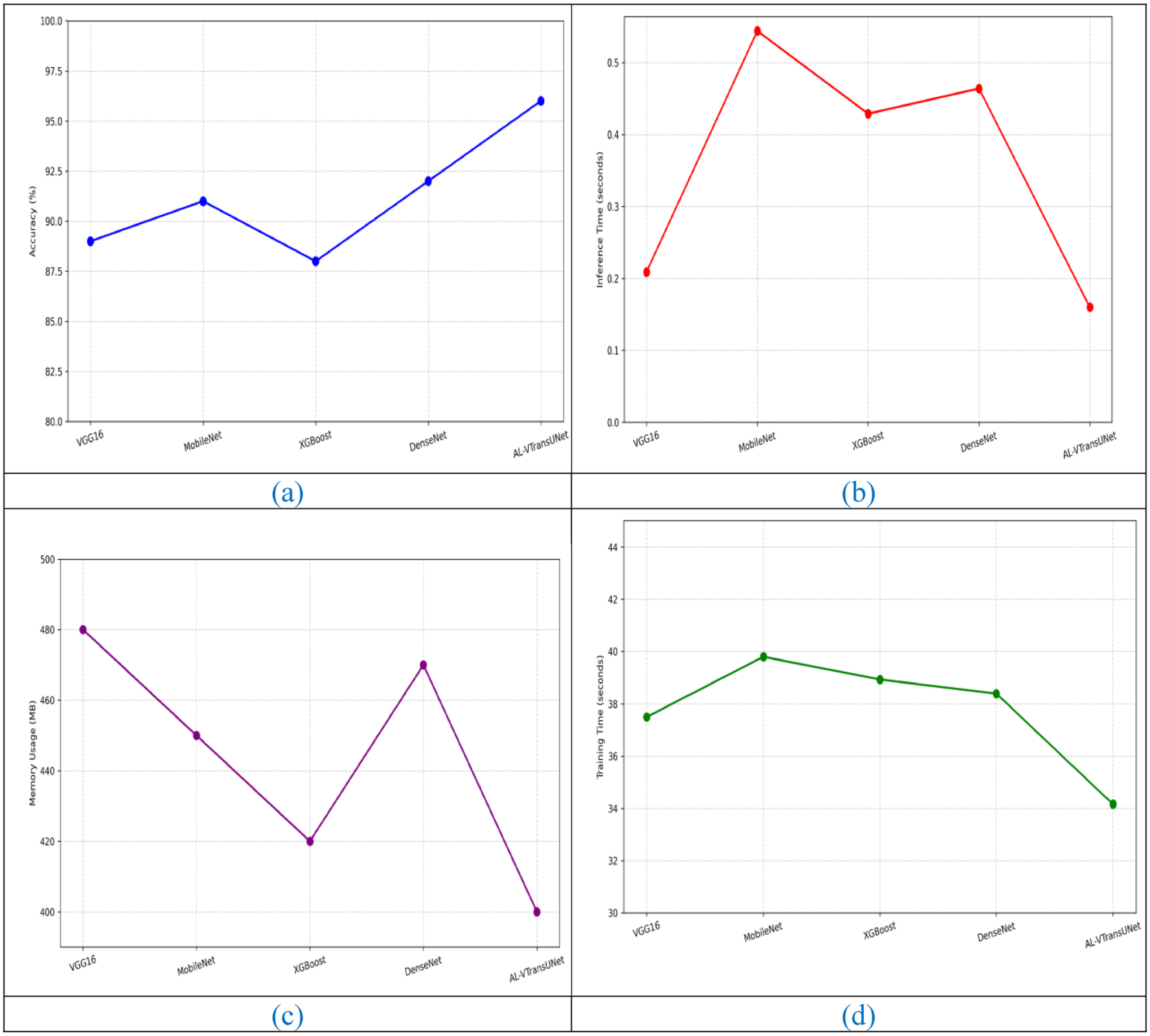


Fig. 18. Analysis of AL-VTransUnet model in terms of (a) Accuracy, (b) Inference Time, (c) Memory usage, and (d) Training Time.

Data availability

The dataset generated and/or analyzed during the current study is not publicly available due to privacy concerns. However, access to the dataset may be granted on a case-by-case basis. Interested researchers can request access by contacting the corresponding author using the contact details provided .

Received: 7 October 2024; Accepted: 19 June 2025
Published online: 30 September 2025

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Author contributions

Deepa J and Dr. Madhavan P designed the model, computational framework and carried out the implementation. Dr. Madhavan P performed the calculations and wrote the manuscript with all the inputs. Deepa J, Dr. Madhavan P discussed the results and contributed to the final manuscript.

Funding

This research did not receive any specific funding.

Competing interests

The authors declare no competing interests.

Additional information

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